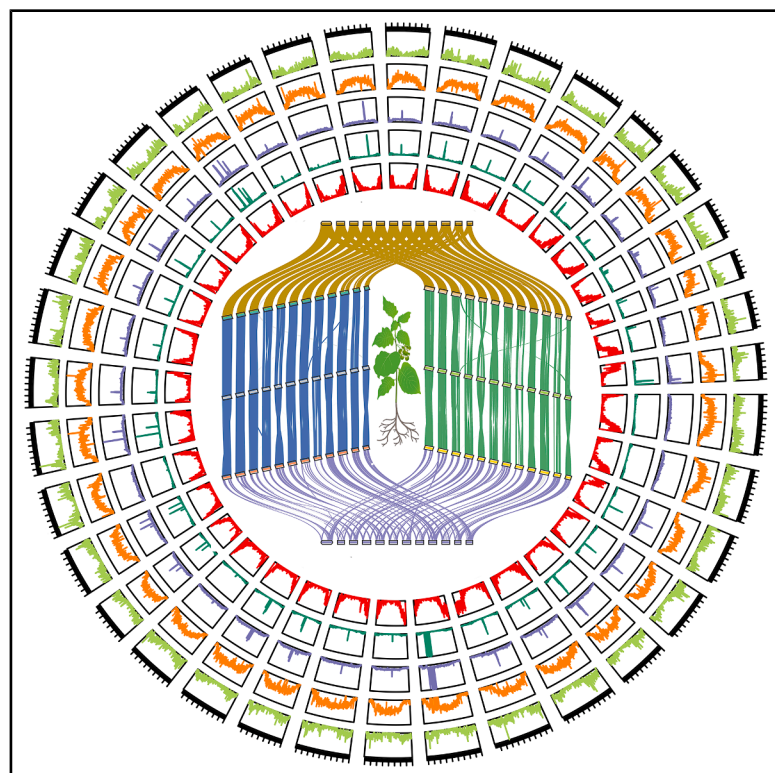


Haplotype-resolved genome of hexaploid *Solanum nigrum*: Insights into origin, evolution, and steroidal glycoalkaloid biosynthesis

Graphical abstract



Authors

Bo Kong, Xiaoxu Li, Zhenchen Zhang,
Wei Luo, Yunpeng Cao

Correspondence

xfcpeng@126.com

In brief

Kong et al. present a haplotype-resolved assembly of the hexaploid *S. nigrum* genome, providing insights into its evolutionary history and significant chromosomal rearrangements. The study identifies key genes related to steroidal glycoalkaloid biosynthesis, paving the way for future research and practical applications within the *Solanum* family.

Highlights

- Haplotype-resolved, chromosome-level genome of hexaploid *Solanum nigrum* is presented
- Hexaploid *S. nigrum* originated through hybridization of tetraploid and diploid progenitors
- 26 core genes for SGA synthesis are identified to facilitate future metabolic bioengineering



Article

Haplotype-resolved genome of hexaploid *Solanum nigrum*: Insights into origin, evolution, and steroidal glycoalkaloid biosynthesis

Bo Kong,^{1,3} Xiaoxu Li,^{1,3} Zhenchen Zhang,¹ Wei Luo,¹ and Yunpeng Cao^{2,4,*}¹Research Institute of Technology Center, China Tobacco Hunan Industrial Co., Ltd., Changsha, China²Guangxi Key Laboratory of Forest Ecology and Conservation, Key Laboratory of National Forestry and Grassland Administration on Cultivation of Fast-Growing Timber in Central South China, School of Forestry, Guangxi University, Nanning 530004, China³These authors contributed equally⁴Lead contact*Correspondence: xfccypeng@126.com<https://doi.org/10.1016/j.celrep.2026.117037>

SUMMARY

Black nightshade (*Solanum nigrum*) is a valuable medicinal plant and emerging model for polyploid evolution, but its study is limited by the lack of a reference genome due to complex polyploidy. We present the first haplotype-resolved, chromosome-level assembly of the 6.1 Gb *S. nigrum* genome, organized into 12 homoeologous groups. We confirm that one diploid parent of *S. nigrum* is *S. americanum* and identify significant chromosomal rearrangements between them. Our analysis indicates that *S. nigrum* originated approximately 8.8 million years ago from hybridization between tetraploid and diploid progenitors followed by whole-genome duplication. Integrating multi-tissue transcriptomics, we identify 65 steroidal glycoalkaloid (SGA) biosynthesis genes that undergo sub-functionalization and neo-functionalization. We also characterize 26 core genes involved in the SGA biosynthetic pathway, establishing a framework for metabolic engineering in heterologous systems. These findings offer crucial genomic resources for advancing research, breeding, and industrial applications of *S. nigrum* and the broader Solanaceae family.

INTRODUCTION

Solanaceae is one of the largest families of Angiosperm plants and includes economically important species such as tomato, potato, and eggplant. In addition to cultivated species, several other species, including *Solanum nigrum*, *Datura stramonium*, and *Atropa belladonna*, have garnered significant attention for their lethal toxicity and potential medicinal properties.^{1,2} Among them, *S. nigrum*, a common perennial plant native to Eurasia, has been introduced to South Africa, Australasia, and America.^{3,4} In many countries, both leaves and ripe fruits of *S. nigrum* have been used for traditional medicine and culinary purposes.³ Previous studies have identified various beneficial compounds in *S. nigrum*, including polyphenolics, glycoalkaloids, glycoproteins, and anthocyanidins, as well as amino acids like valine, serine, L-proline, isoleucine, alanine, and aspartic acid, suggesting its potential as a beneficial food source. Nevertheless, solanine, a toxic steroidal glycoalkaloid (SGA), is present in various parts of *S. nigrum*, with the highest levels being found in green unripe fruits and young leaves. Only the ripe fruits or cooked leaves of *S. nigrum* should be consumed to prevent toxicity.

Recent genome editing techniques like CRISPR-Cas9 and transcription activator-like effector nuclease (TALEN) are powerful tools for crop improvement and engineering of metabolites that can be used as medicinal compounds in Solanaceae. For

instance, genome-edited plants utilizing TALEN successfully decreased the level of cholesterol and SGAs like toxic α -chaconine and α -solanine in potato.⁵ Four *FucT* genes and two *XyIT* genes with a total of 12 alleles altogether were knocked out using CRISPR-Cas9⁶ and TALEN⁷ in *Nicotiana benthamiana*, whereas in *N. tabacum*, the same genes were targeted using CRISPR-Cas9⁸ to create pharmacological protein and antibody expression in plants. The onset of fruit ripening in *slnam1*-deficient mutants created by CRISPR-Cas9 technology was delayed compared with the wild type in tomato.⁹ The success of these methods in these Solanaceous species is attributed to the availability of high-quality genome data. In contrast, no successful and efficient methods for generating genome-edited plants in *S. nigrum*, an allopolyploid ($2n = 72$) with a complex genome (AABBBB), have been reported.^{10,11}

Steroidal alkaloids (SAs) and their glycosylated derivatives, SGAs, are nitrogenous toxins abundant in Solanaceous plants such as *S. lycopersicum*, *S. tuberosum*, *S. melongena*, and *S. nigrum*.^{12–17} SGAs are synthesized from cholesterol through sequential reactions catalyzed by cellulose synthase-like, cytochrome P450s (CYPs), dioxygenases, and UDP-glucosyltransferases (UGTs).^{12–16,18} Among them, UGT-mediated terminal glycosylation of GAME1, GAME17, GAME18, and GAME2 is pivotal for controlling SGA solubility, stability, and bioactivity. Although these genes are well characterized in some



Solanaceous crops, the SGA pathway has not yet been reconstructed in model plants. Here, we employ high-fidelity long reads (HiFi) and high-throughput chromosomal conformation capture (Hi-C) to generate the first haplotype-resolved, chromosome-level genome of *S. nigrum*, providing a vital foundation for dissecting SGA biosynthesis. Comparative genomics of hexaploid *S. nigrum* and its diploid ancestors elucidated the history of genome polyploidization and demonstrated that the current hexaploid resulted from auto-allopolyploidization. The genomic resources and analyses provided here are essential for understanding Solanaceae evolution and serve as valuable tools for future molecular breeding, biological research, and industrial applications of *S. nigrum*.

RESULTS

Assembly of a highly contiguous and phased hexaploid reference genome

S. nigrum ($2n = 6x = 72$), commonly referred to as black nightshade, is a widely distributed species within the *Solanaceae* family, found across diverse regions including Europe, Asia, North America, and Africa (Figure 1A). Given the high heterozygosity of *S. nigrum*, it is crucial for genetic breeding and functional research to obtain a haplotype-resolved assembly that captures all gene allele variations and homologs, similar to those in autotetraploid potatoes (*S. tuberosum*) and alfalfa (*Medicago sativa*). We selected a local cultivar of *S. nigrum* with $2n = 6x = 72$ chromosomes for genome sequencing and generated 133.9 Gb Illumina short reads (Table S1). K-mer analysis estimated the haploid genome size at approximately 3.0 Gb, with a genome survey indicating that *S. nigrum* has a high repeat content of 64.13%–64.15% (Figure S1 and Table S2). An independent flow cytometry measurement reported a larger genome size of 3.767 Gb,¹¹ a discrepancy that is well-explained by the high repetitive content (64.13%) of the *S. nigrum* genome, which is challenging to fully assemble but is included in physical DNA measurements. The 276.24 Gb clean HiFi reads were assembled into two contig-level haplotype assemblies with N50 values of 79.65 and 81.25 Mb and assembly sizes of 3.09 and 3.01 Gb (i.e., SnGA_v0.1 assembly, Table S3), which are similar to the estimated genome size. Subsequently, we employed genomic Hi-C sequencing to obtain the spatial organization data for 72 *S. nigrum* chromosomes, facilitating the assembly of contigs into haplotype-resolved chromosome-level scaffolds (Figure 1B). Using Hi-C read pairs, the final 72 chromosome-level haplotype-resolved scaffolds were obtained through two rounds of the divide-and-conquer technique, followed by a modest amount of manual curation in Juicebox. Subsequently, the contigs and HiFi reads were first used to fill the gaps in the chromosome-level genome using the GapFiller module in the quArTeT software.¹⁹ The remaining gaps were subsequently filled using an innovative cyclic elongation technique called DEGAP. Finally, following the removal of plasmids and redundant sequences, telomere patching, and repeat-aware polishing with HiFi and Illumina reads, we achieved a nearly complete reference genome consisting of 72 pseudo-chromosomes totaling 6.1 Gb (Table 1).

To further support comparative studies, a chromosome-level assembly of the diploid *S. americanum* ($2n = 2x = 24$), a possible

ancestral species to *S. nigrum*, has been generated using a similar hybrid assembly method. The genome assembly successfully mapped 1.04 Gb of contigs onto 12 pseudochromosomes. The contig N50 length of 42.87 Mb and scaffold N50 length of 87.38 Mb indicate a significantly improved genome assembly (Table 1), representing a substantial advancement over the previous scaffold-level assembly of the *S. americanum* genome.²⁰ Intergenomic comparison analysis revealed the anticipated 3-to-1 syntenic relationship between *S. nigrum* haplotype-genome and *S. americanum* genome, with some regions exhibiting translocations or inversions (Figures 1D and S2). Large-scale inter-chromosomal translocations were nearly absent, and structural rearrangements were primarily intra-chromosomal. These included at least six large inversions on Chr01, Chr03, Chr04, Chr06, and Chr10, and more complex rearrangements on chr5, chr9, and chr11. In contrast, chr8 was the most structurally conserved, while chr2, chr7, and chr12 were also relatively conserved, showing few collinearity breaks (Figure S2).

Multiple independent assessments were conducted to evaluate the correctness and completeness of the *S. nigrum* genome. First, data computations indicate that the homologous chromosomal groups in *S. nigrum* are almost six times larger than those in the diploid genome of *S. americanum* and the published diploid genome *S. rostratum*.²¹ Then, the assembled genome was aligned with HiFi reads and Illumina short reads, achieving mapping rates of 97.75% and 98.76%, respectively, which confirmed the accuracy and completeness of the sequence assembly. Benchmarking universal single-copy orthologs (BUSCO) analysis identified 99.8% of eukaryotic conserved genes in the chromosome-scale genome of *S. nigrum*, indicating a high completeness of the genome assembly (Tables 1 and S4). Merqury analysis of the *S. nigrum* genome assembly using HiFi data yielded an average quality value (QV) of 59.63, corresponding to a base error rate of less than 10^{-6} . Following the annotation of long terminal repeats (LTRs), the genome achieved an LTR assembly index (LAI) score of 20.01, closely aligning with the scores of the autotetraploid *M. sativa* (22.30) and allohexaploid *Echinochloa colona* (22.5), thereby achieving the gold standard for reference genomes. Collectively, these results demonstrate a high degree of continuity and completeness in the *S. nigrum* genome.

Genome annotation

Using transcriptome data, homology-based prediction, along with *de novo* prediction, we identified 177,059 protein-coding genes across six *S. nigrum* haplotype genomes (Table 1), a significantly higher number compared to other *Solanum* species, which range from 28,154 in *S. rostratum*²¹ to 152,855 in *S. tuberosum*.¹⁷ The overall BUSCO completeness score of 95% across the six *S. nigrum* haplotype genomes is highly comparable to the annotations of the *S. rostratum*, *S. tuberosum*, and *S. americanum* assemblies (Figure 2A and Table S5). As anticipated, the genes were distributed unevenly, with a higher concentration toward the ends of the chromosomal arms. Approximately 99.80% of the predicted protein-coding genes were functionally annotated through searches across multiple databases, including Pfam (99.70%), Kyoto Encyclopedia of Genes and Genomes (KEGG) (99.50%), and Gene Ontology (GO) (99.49%) (Figure 2A and Data S1).

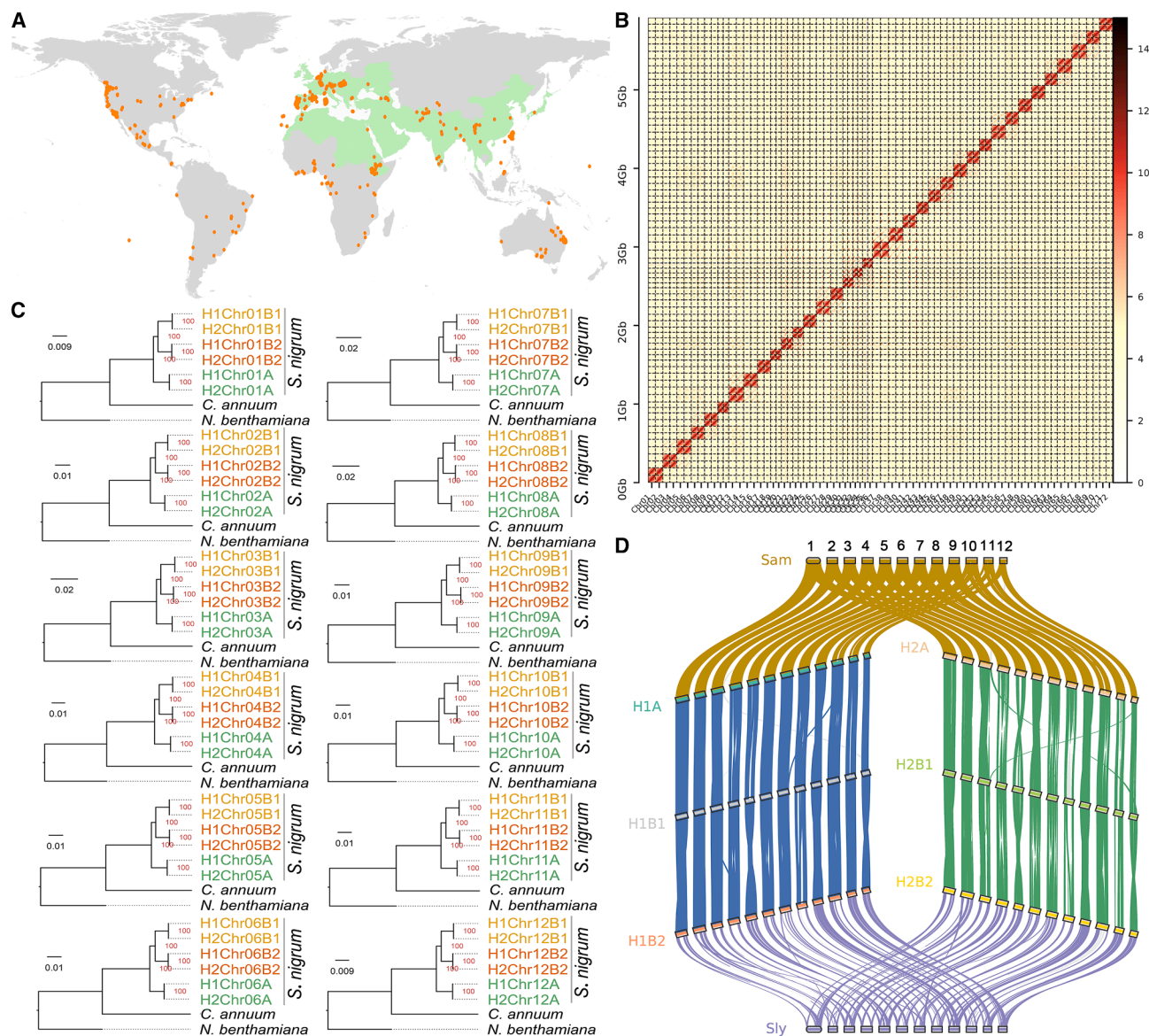


Figure 1. The biological profile and genome of *S. nigrum*

(A) The ornamental hybrid *S. nigrum* is highlighted in orange, whereas the natural distribution of the extant species is shown in green. The natural distribution range (green) is based on POWO (Plants of the World Online; <https://powo.science.kew.org/taxon/urn:lsid:ipni.org:names:30048260-2>), while the orange dots show occurrence data compiled from GBIF (Global Biodiversity Information Facility; accessed on March 5, 2025).

(B) Heatmap illustrating genomic interactions among *S. nigrum* chromosomes.

(C) Phylogeny-based genomic phasing of *S. nigrum*, with red, orange, and green branches representing subgenomes A, B1, and B2, respectively.

(D) Intragenomic synteny of various *S. nigrum* subgenomes alongside *S. americanum* and *S. lycopersicum*.

In our analysis of the assembled genome of *S. nigrum*, we identified a significant presence of transposable elements (TEs), amounting to 4,389 Mb, which is 72.01% of the total genomic assembly (Figure 2A and Table S7). This proportion is comparable to that observed in closely related species, such as *S. americanum* (72.80%, 760.6 Mb) and *S. rostratum* (72.26%; Zhang et al.^{1,21}, 2023), as well as *S. pimpinellifolium* (67.3%; Wang et al.²). Notably, LTR retrotransposons dominate the composition of the *S. nigrum* genome, comprising 60.23% of

the total, with specific subcategories including 4.89% attributed to Ty1-copia and 39.34% to Ty3-gypsy. In comparison, *S. americanum* reflects a similar trend, with LTRs accounting for approximately 60.61% of its genome, including 5.21% Ty1-copia and 39.37% Ty3-gypsy. Tandem repeats, including microsatellites and minisatellites, comprise another 10.58% of the genome (Table S6). These findings highlight the significant role of TEs in shaping the genomic architecture of these species within the *Solanum* genus (Figure 2A and Table S7).

Table 1. Genome assembly statistics for *Solanum nigrum* and *S. americanum*

Genomic features	<i>S. nigrum</i>	<i>S. americanum</i>
Ploidy	6	2
Total assembly size of contigs (Gb)	6.1	1.04
Number of contigs	2570	104
Maximum contig length (Mb)	118.63	82.99
N50 contig length (Mb)	79.76	42.87
Number of scaffolds	2526	61
Number of chromosomes	72	12
N50 scaffold length (Mb)	84.11	87.38
Number of gaps	5	22
BUSCO completeness of assembly (%)	99.8	99.3
Repetitive sequences (%)	72.01	72.8
Number of annotated genes	130,186	25,971
BUSCO completeness of annotation (%)	95	95.3
Quality value (QV)	59.63	–
LTR assembly index (LAI)	20.01	–

In polyploid genomes, alleles are defined by homologous genes at the same locus.^{22,23} In our study, we annotated 28,575 genes with distinct alleles: 47.64% (13,613 genes) had six alleles (Table S8), 3.55% (1,015 genes) had five, 18.53% (5,295 genes) had four, 2.93% (836 genes) had three, and 21.81% (6,233 genes) had two, while 5.54% (1,583 genes) had a single allele. This resulted in a total of 124,490 alleles, averaging 4.62 alleles per gene. Additionally, we identified 4,596 genes with signs of dispersed duplication and 422 genes with tandem duplication (Table S8).

Origin of *S. nigrum* from hybridization between a tetraploid and a diploid *Solanum* species

In the newly formed hexaploids, such as *S. nigrum* and *Helianthus tuberosus*, which exhibit high heterozygosity (Table S2), elucidating homologous chromosome relationships is critical for understanding genome structure and polyploid origin. During meiosis, the six chromosomes within a homologous group typically form three heterozygous synaptic pairs. A distinct subgenome structure is inferred if inter-pair divergence substantially exceeds intra-pair variation. Utilizing a haplotype-resolved, chromosome-scale assembly of *S. nigrum*, we examined sequence divergence across all 12 homologous groups. Our initial comparisons indicated nearly uniform divergence (~2%) among all six chromosomes within each group, obscuring synaptic pair discrimination. We attributed this ambiguity to potential noise from whole-chromosome alignments. To refine our analysis, we focused on single-copy genes within each homologous group. We reconstructed the phylogeny of homologous chromosomes using concatenated multiple sequence alignments (MSAs) of single-copy genes from 12 homologous groups. The resulting topology and branch lengths showed that the hexaploid *S. nigrum* is an autoallopolyploid with three subgenomes: A, B1, and B2 (Figure 1C), a finding also supported

by Smudgeplot (Figure 3A) and K-based clustering analysis (Figure S3).

The origin of hexaploid *S. nigrum* remains unverified due to insufficient high-quality genomes for whole-genome alignment. Our evolutionary and Smudgeplot analyses suggest that the direct ancestors of *S. nigrum* are likely a diploid and an autotetraploid species (Figure 3). The hybridization of these ancestors, followed by chromosome doubling, likely resulted in the speciation of *S. nigrum*. Similarly, previous studies on *S. nigrum* based on its cytological research also believed that its hexaploid genome structure is ABB,¹⁰ with *S. americanum* as a possible close extant relative of an ancestor.^{20,24,25} Our study provides strong genome-scale evidence supporting the hybrid origin of auto-allopolyploid *S. nigrum* (Figure 3). To further validate these findings, we constructed a low-copy maximum-likelihood (ML) tree for sixteen diploid species: *S. didymium*, *S. aciniatum*, *S. guineense*, *S. seaforthianum*, *S. dulcamara*, *S. physalifolium*, *S. ptychanthum*, *S. americanum*, *S. paucidentis*, *S. enantiophyllanthum*, *S. macrotonum*, *S. nigrescens*, *S. furcatum*, *S. douglasii*, *S. sinuatirecurum*, and *S. longifilamentum*. This tree showed that *S. americanum* clustered with *S. nigrum* B1 and B2, supporting the hypothesis that it is a close extant relative of an ancestor of the hexaploid *S. nigrum* (Figure S4). Remarkably, we also quantified the genome-wide sequence divergence between each *S. nigrum* subgenome (A, B1, and B2) and three closely related diploid *Solanum* species (*S. americanum*, *S. paucidentis*, and *S. ptychanthum*) using one-to-one orthologous genes. For both B1 and B2, *S. americanum* showed the lowest median Ks values among all sampled diploids (Figure S5), strongly supporting that it represents the closest extant relative of the progenitor of these subgenomes. Overall, these Ks-based genomic proximity analyses align with the low-copy nuclear phylogeny and together support the origin of hexaploid *S. nigrum* from a B-like diploid ancestor most closely related to *S. americanum*.

We treated each subgenome as a virtual species and analyzed the divergence times between them and closely related species. Our results indicated that *S. seaforthianum* diverged from its relatives between 8.8 and 14.65 million years ago (MYA), with inter-specific hybridization and chromosome doubling occurring between 3.5 and 7.01 MYA, resulting in the emergence of hexaploid *S. nigrum* (Figure 3B). Based on time tree estimates, we traced the evolution of *S. nigrum* from a diploid ancestor to a hexaploid. Approximately 8.8 MYA, two perennial diploids (AA and BB) diverged from this ancestor. In the diploid B genome donor, unreduced gamete fusion or somatic chromosome doubling produced a tetraploid (BBBB). Subsequently, the diploid gamete (BB) combined with a haploid gamete (A) to form a triploid hybrid (ABB), which eventually underwent chromosome doubling to become the original hexaploid (AABBBB) *S. nigrum* (Figure 3C). Over millions of years of variation, evolution, vegetative reproduction, and domestication, this hexaploid evolved into the current highly heterozygous *S. nigrum*. In our study, we employed haplotype H1 for subsequent analyses unless specified otherwise.

Comparative genomics and whole-genome duplication in *S. nigrum* genome

OrthoFinder grouped 95.90% (694,583) of the 724,253 genes from 20 species into 35,430 orthologous groups (OGs). OG analysis

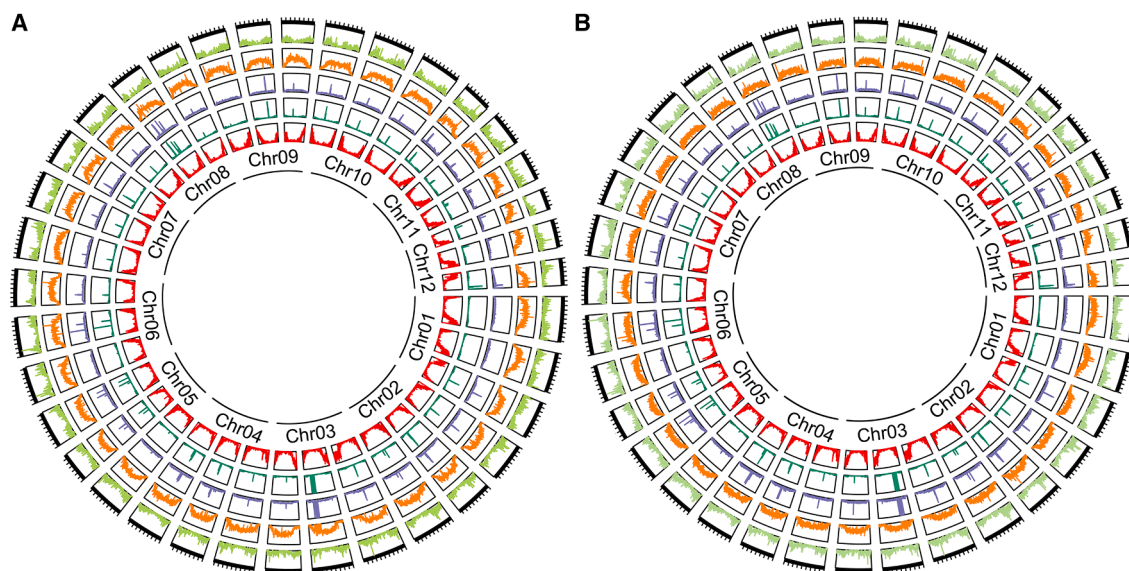


Figure 2. Landscape of genome assembly and annotation for the *S. nigrum* genome

Basic genomic components are distributed across the *S. nigrum* haplotypes 1 (A) and 2 (B), including TIR transposon density, LTR retrotransposon density, Gypsy LTR retrotransposon density, Copia LTR retrotransposon density, and gene density, from inner to outer tracks.

revealed that the 5,709 largest OGs, each containing ≥ 28 genes ($G50 = 28$, $O50 = 5,709$), accounted for 50% of the genes (Table S9). Among these OGs, 4,987 were species-specific (containing 24,259 genes), and 7,581 were present in all species, including 308 OGs composed solely of single-copy genes (Table S9). Gene family expansion/contraction analysis showed expansion in 285, 285, and 351 gene families in the A, B1, and B2 subgenomes of *S. nigrum*, respectively. While the extent of expansion was less than in its parent species, *S. americanum* (791 gene families), the degree of genome contraction was substantially greater (Figure 3B).

Phylogenetic analysis of single-copy orthologs using MCMCTree estimated the divergence between *Solanum* and non-*Solanum* plants at 15.87–22.26 MYA. *S. nigrum* and *S. seaforthianum* are estimated to have diverged 8.80–14.65 MYA (Figure 3B). Whole-genome duplication (WGD) events have frequently occurred in flowering plant evolution, significantly influencing genome evolution.^{26–28} Chromosome-scale synteny, karyotype analysis, Smudgeplot analysis, and K-means clustering independently and strongly support the hexaploid origin of *S. nigrum*. These converging lines of evidence demonstrate that *S. nigrum* is a hexaploid, not a simple diploid, resulting from a complex polyploidization event. Indeed, analysis of synonymous substitution rates (K_s) for homologous gene pairs within syntenic regions of the *S. nigrum* genome indicates that it experienced an ancient whole-genome triplication event (WGT- γ) (Figures 4A, S6–S8), which is widespread in core eudicots.²⁹ Furthermore, the study identified a Solanaceae-specific WGD event, with K_s values ranging from 0.70 to 0.77 (Figures 4A and S6–S13), estimated to have occurred approximately 41.99–50.44 MYA (Figure 4A). Further studies revealed that the peak of K_s distribution between paralogs of subgenomes A and B1 was nearly identical to that between paralogs of subgenomes A and B2 ($K_s = 0.673$) (Figure 4B).

In contrast, the peak between paralogs of subgenomes B1 and B2 was different ($K_s = 0.664$) from those between subgenomes A and B1 and subgenomes A and B2 (Figure 4B). This suggests that subgenome B from the diploid ancestor of *S. americanum* rapidly underwent homologous doubling to form the tetraploid B1B2, followed by a swift transition to hexaploidy, which included the diploid ancestor of subgenome A.

Murat et al. (2017) proposed that the hypothetical ancestral eudicots karyotype (AEK) comprised 7 ancestral, pre-duplication chromosomes.³⁰ We investigated the potential karyotype changes in *S. nigrum* and other Solanaceae species using the hypothetical AEK model developed by Sun et al. (2022),³¹ which comprises 23,374 ancestral genes (genes before ancestral duplication). Karyotype analysis indicates that chromosome 11 of subgenome B experienced the most extensive fusion and fission events, incorporating fragments from six of the seven ancestral chromosomes (Figure 4C). In contrast, chromosomes 5 and 6 exhibited minimal rearrangement, primarily originating from recombination of AEK2/AEK4 and AEK6/AEK7 (Figure 4C), respectively, aligning with findings in other Solanaceae. This suggests a conserved and stable role for chromosomes 5 and 6 within the Solanaceae genome. The three subgenomes exhibit high karyotype similarity, particularly between B1 and B2, which are nearly identical and closely resemble the karyotype of *S. americanum*, supporting the hypothesis that *S. americanum* is the B subgenome donor for hexaploid *S. nigrum*. The primary karyotype differences between subgenomes A and B involve large-scale rearrangements of chromosomes 3 and 12, specifically a translocation (Figure 4D).

Homoeolog expression patterns in *S. nigrum* genome

To determine patterns of homoeolog expression, we focused on 41,847 expressed genes (13,949 syntenic triads in total 13,891

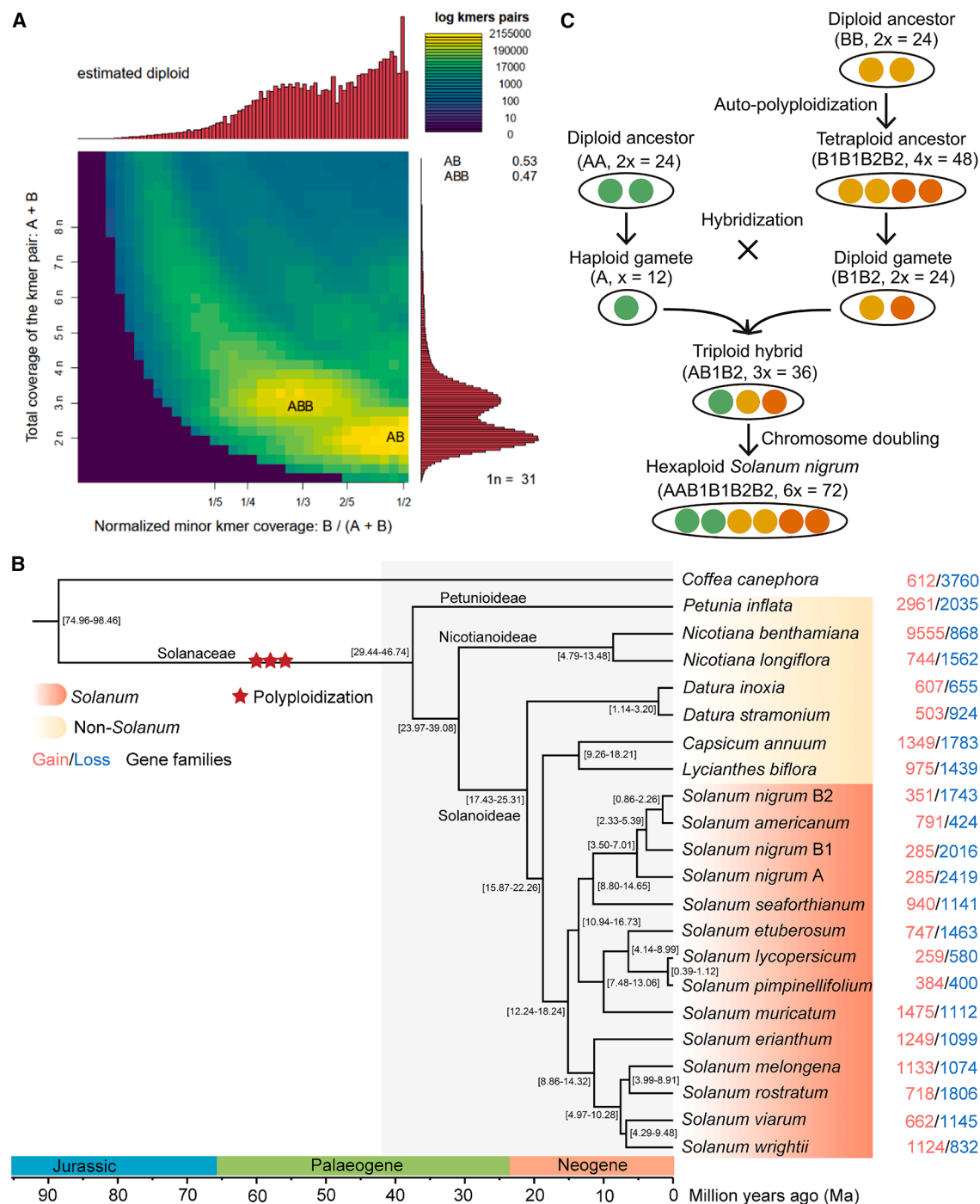


Figure 3. Formation of auto-allohexaploid *S. nigrum* through hybridization

(A) Smudgeplot analysis based on kmers for *S. nigrum*.

(B) Divergence time tree of Solanaceae plants, with coffee (*C. canephora*) as the outgroup.

(C) Diagram illustrating the lineage from the diploid ancestors (AA and BB) and tetraploid ancestor (B1B1B2B2) leading to chromosome doubling and hybridization events that resulted in hexaploid *S. nigrum* (AAB1B1B2B2).

triads, 95.99%) that had a 1:1:1 gene correspondence across the three haploid assemblies and a homoeolog expression of >0.5 transcripts per million (TPM) in at least one tissue. We first

standardized the relative expression of A, B1, and B2 homoeologs for each tissue and combined analysis for all 41,847 expressed triads, followed by a global analysis integrating data

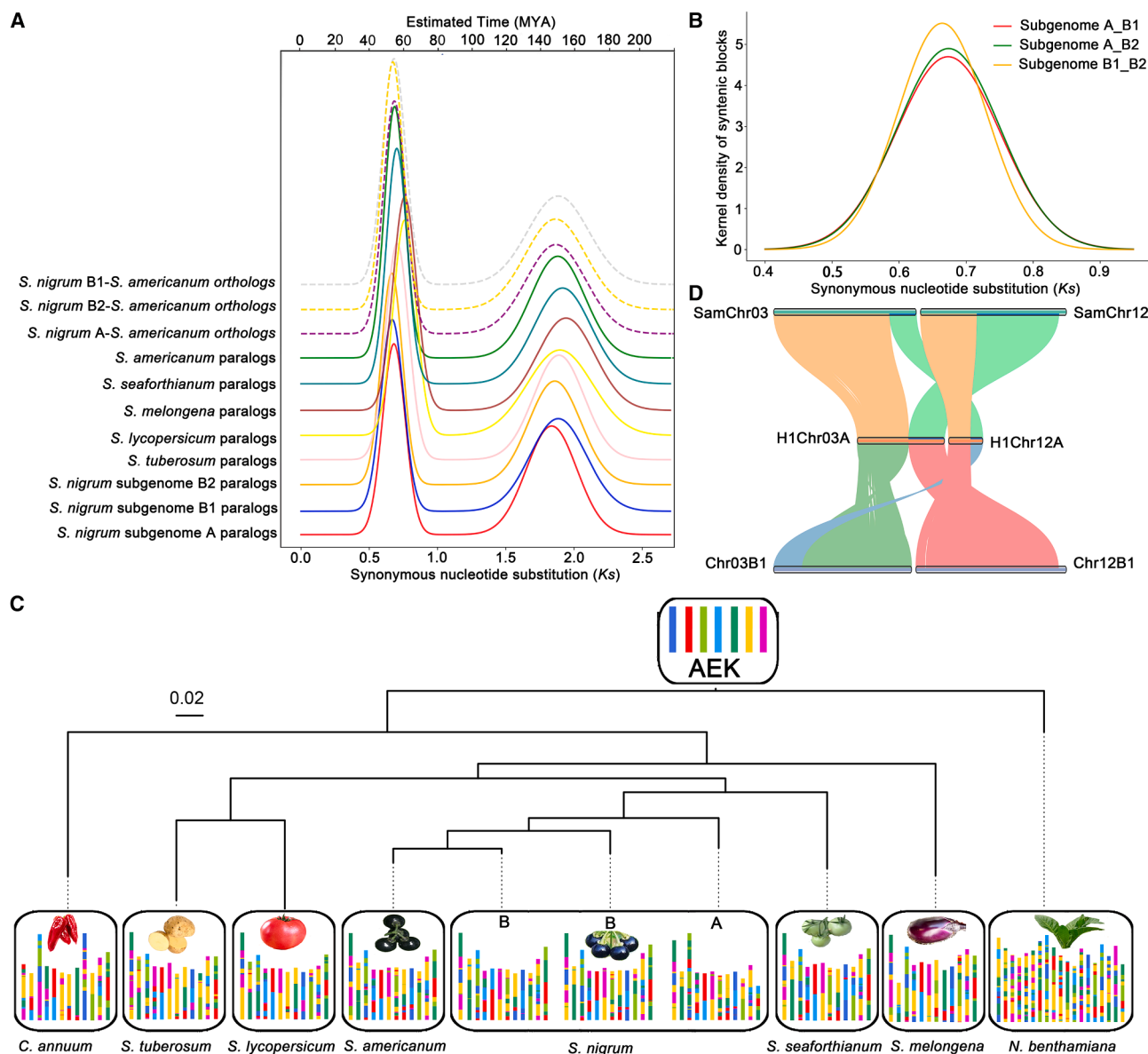


Figure 4. Analysis of karyotype evolution and WGD in *S. nigrum*

(A) WGD analysis based on Ks distribution, showing the Ks distribution for orthologs (dashed lines) and paralogs (solid lines) in collinear areas of *Solanum*.

(B) Ks distribution of orthologs among the three subgenomes in collinear regions.

(C) The rebuilt AEK model forecasts the evolution of sister-to-core eudicot karyotypes, with different colors representing the seven protochromosomes in the hypothetical AEK.

(D) A detailed view of the syntentic connections between chromosomes 3–12 of *S. nigrum* in the A, B1, and B2 sub-genomes.

from all twelve tissues or developmental stages. On average, ~40% of homoeolog triads showed non-balanced expression patterns, with higher (homoeolog-dominant) or lower expression (homoeolog-suppressed) from a single homoeolog respecting the other two (Figure 5A). As for each tissue, balanced triads ranged from 62.30% in the seed to 77.23% in the fruit. The homoeolog-dominant triads were found at moderate frequencies for each tissue (ranging from 0.50% to 4.91%), whereas the homoeolog-suppressed triads were more frequent (from 2.71% to

9.59%) (Figures 5B and 5C). Genes from syntenic triads in the dominant groups had higher absolute transcript abundance than genes in the balanced or suppressed groups (Figures 5B and 5C).

To further investigate the global landscape of homoeolog expression regulation, we performed principal component analysis (PCA) and hierarchical clustering on expression bias across all twelve tissues (Figures S14–15). PCA revealed clear tissue-specific clustering patterns, with PC1 (28.9%) primarily

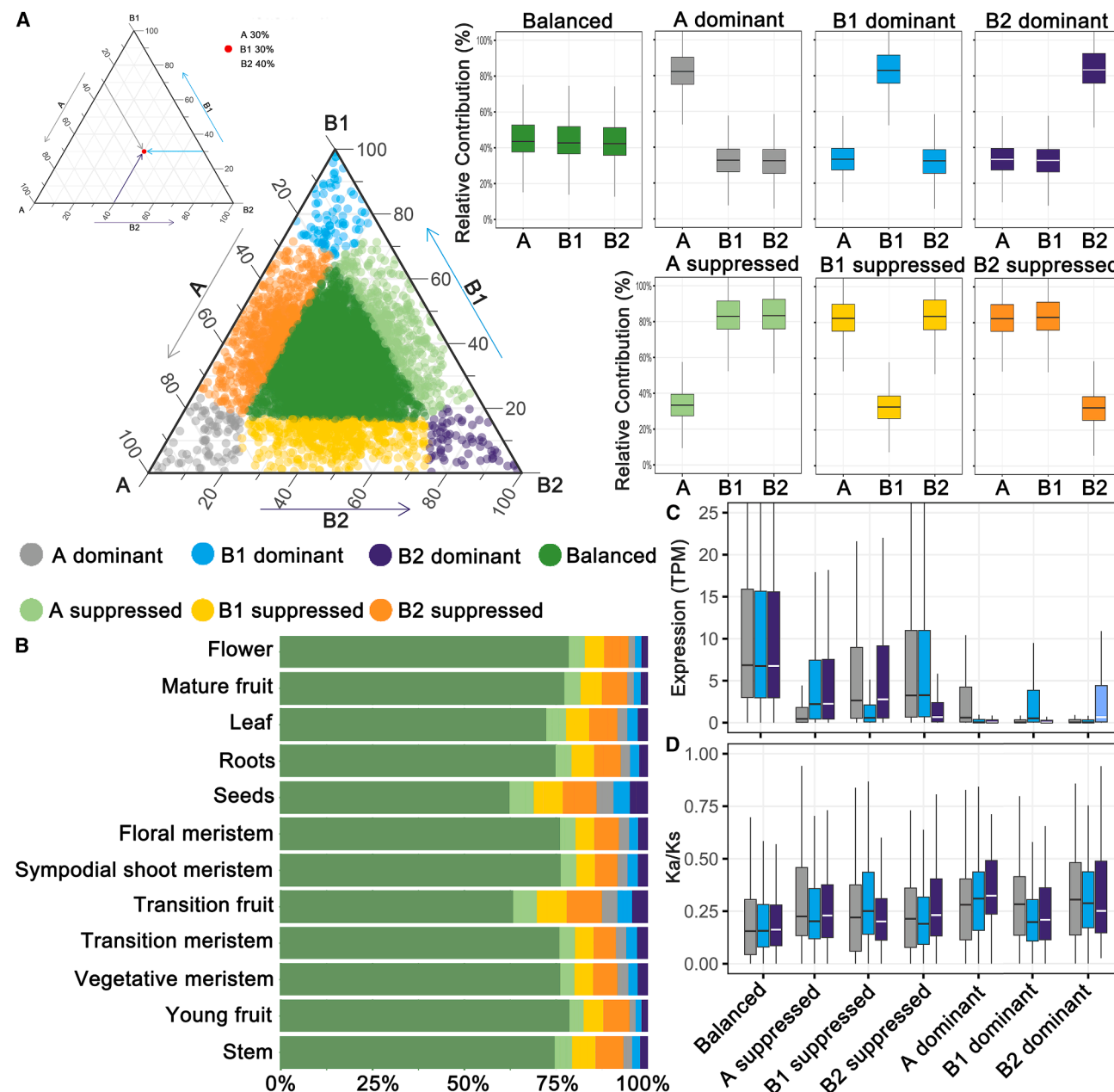


Figure 5. Bias in homoeolog expression within syntenic homoeolog triads

(A) A ternary plot shows the relative expression levels of syntenic triads in the *S. nigrum* haplotype 1 genome. The A, B1, and B2 coordinates reflect each homoeolog's contribution to total triad expression, with each circle representing a gene triad (illustrated in the upper left). Homoeolog-dominant groups are indicated by colored circles at the vertices, homoeolog-suppressed groups by circles near the edges and between vertices, and balanced triads by green circles. Boxplots illustrate the relative contribution of each subgenome based on triad assignments to the seven categories.

(B) The percentage of triads in each homoeolog expression bias group across all 12 tissues.

(C) Boxplots of absolute TPM expression abundance for the seven subgenomes.

(D) Violin plots of Ka/Ks values across the seven categories. The balancing category shows slightly lower Ka/Ks values compared to the homoeolog-dominant and homoeolog-suppressed categories.

separating vegetative tissues (leaf, stem, and root) and meristems (VM, vegetative meristem; TM, transition meristem; and SYM, sympodial shoot meristem) from reproductive and fruit developmental tissues (flower, seed, TF, transition fruit; YF,

young fruit; and MF, mature fruit), while PC2 (23.5%) captured variation among developmental stages within these categories (Figure S14). Notably, the three subgenomes (A, B1, and B2) from the same tissue clustered together rather than forming

subgenome-specific groups, indicating that tissue identity is the primary driver of homeolog expression bias rather than inherent subgenome dominance. Hierarchical clustering analysis further revealed distinct gene modules with coordinated expression bias patterns across tissues (Figure S15). Several clusters exhibited tissue-specific subgenome switching, where B1-dominant expression in vegetative tissues and meristems transitioned to A-dominant or balanced expression in reproductive tissues, suggesting sub-functionalization following polyploidization. The fruit developmental series (TF, YF, and MF) and meristem developmental progression (VM, TM, FM [floral meristem], and SYM) showed gradual shifts in homeolog bias along PC axes, indicating dynamic regulation of subgenome expression during fruit maturation and meristem differentiation.

Heterozygosity differences between subgenome pairs can indicate subgenome dominance, with recessive subgenomes expected to face less purifying selection pressure than dominant ones.³² In our study, the non-synonymous (Ka)/Ks ratios of non-balanced triad genes (homeolog-dominant or homeolog-suppressed) are typically higher than those of balanced triads (Figure 5D), indicating that non-balanced triad genes may have undergone stronger positive selection and potentially acquired significant adaptive advantages during evolution.

Fractionation and neo-/sub-functionalization in *S. nigrum* genome

Comparative analysis of homologous genes is crucial for understanding gene function and evolution in plant genome research.³² Analysis of homeologous gene composition across the three subgenomes revealed that more than two-thirds of the genes (42,697 out of 62,142) are shared, with many of these genes enriched in functions associated with obsolete cells, intracellular fractions, and plastid fractions, highlighting their functional redundancy (Figure S16). The remaining shared genes were enriched in hormone and signal transduction, plant growth, and stress responses, demonstrating that key biological functions are highly conserved despite genomic variations and recombination. In the pseudohaploid subgenomes, about one-third of the genes (19,467 out of 62,142) lacked homeologs in one or two of the remaining subgenomes, suggesting that redundant copies of a large number of genes were selectively eliminated after the WGD events (i.e., fractionation). Subgenome A has 3,961 specific genes enriched in DNA repair, cell wall formation, and homologous recombination (Figure S17). Subgenome B1 contains 1,913 specific genes enriched in RNA modification, hydrolase activity, and nucleic acid binding (Figure S18), while subgenome B2 comprises 1,511 specific genes enriched in catalytic activity, secondary metabolic processes, and glucosyltransferase activity (Figure S19). These findings suggest that all three subgenomes have experienced significant fractionation.

We analyzed the expression patterns of retained triplicate genes in *S. nigrum* using transcriptome data from various tissues and fruit development stages to assess if genes retained in triplicate (without fractionation) may have undergone potential sub-/neo-functionalization. Among the 12,333 gene sets retained in triplicate (36,999 genes), 11,100 (90.00%) exhibited significant shifts in expression patterns among gene copies, implying that these genes are subjected to sub-/neo-functionalization. GO

enrichment analysis (corrected $p \leq 0.05$) revealed that sub-/neo-functionalization gene duplicates are enriched in chloroplast and plastid functions, protein localization and transport, and photosynthetic membranes (Figure S20). This suggests that gene duplication maintains photosynthetic organelle structure, regulates photosynthesis, and promotes plant adaptation through functional divergence. Thus, duplicated genes in the *S. nigrum* genome likely underwent extensive sub-/neofunctionalization, contributing to shaping its phenotype.

In silico analyses of SGA biosynthesis-related genes in *S. nigrum*

Plants generate over 200,000 phytochemicals through specialized biosynthetic genes whose enzymes perform crucial oxygenase-mediated oxidation/hydroxylation and UGT-mediated glycosylation.³³ In Solanaceae, the biosynthesis of steroidal alkaloids starts with CYP monooxygenases that selectively oxidize the steroid core. Next, 2-oxoglutarate-dependent dioxygenases (DOX) add secondary hydroxyl groups or induce cyclization, after which UGTs glycosylate the oxidized intermediates to yield bioactive derivatives. *S. nigrum* possesses higher levels of SGAs than other Solanaceae, making it an ideal model for studying their biosynthesis. Its three subgenomes likely contain multiple copies of genes involved in SGA biosynthesis, which are of significant importance. Utilizing the sequences of these three subgenomes, we comprehensively identified all gene family members involved in SGA biosynthesis in *S. nigrum* and other Solanaceae genomes. The results showed that 368 DOX members, 855 CYP members, and 412 UGT members were identified in *S. nigrum* (Figure 6A and Data S2–S4). As a hexaploid species, *S. nigrum* typically presents three homologous copies of members in these gene families (Figure 6A). One or more gene family members from other Solanaceae species form a tight tripartite cluster with these three *S. nigrum* homologs, suggesting coevolution and functional conservation after polyploidization (Figure 6A).

To identify the candidates for SGA-related genes in *S. nigrum* at the genome level, we used functionally validated sequences from tomato, eggplant, and potato as described in previous studies^{12,16,34,36–39} and identified these CYP, DOX, and UGT families to locate their orthologs in *S. nigrum* (Data S5). A total of 65 genes were identified as potential contributors to SGA biosynthesis in *S. nigrum*, including 6 *GAME4* (CYP88B1), 3 *GAME6* (CYP72A188), 5 *GAME8* (CYP72A208), 4 *GAME11* (DOX), 9 *GAME12* (GABAT, γ -aminobutyric acidtransaminase), 3 *GAME15* (CsIG, cellulose synthase-like G), 11 *MAT* (malonyltransferase), 2 *UGT73DU3*, 9 *UGT73L14*, 9 *UGT73L16*, 3 *UGT93M3*, and 10 *UGT93N4* genes (Figures 6A and 6B and Data S6). We focused on the *S. nigrum* genome to examine the expansion patterns of genes related to the SGAs' biosynthesis, which allowed us to infer their evolutionary trajectory during polyploidization. Chromosome analysis revealed that SGA-related genes are primarily concentrated at the chromosome ends, suggesting a crucial role for telomere-adjacent regions in SGA genesis and regulation (Figure 6C and Data S7). Compared to the diploid progenitors, the majority of SGA members in hexaploid *S. nigrum* showed a ratio of approximately 3:1 in gene copy number relative to their diploid counterparts, except for *GAME11*, *MAT*, *UGT93N4*, and *UGT73DU3*.

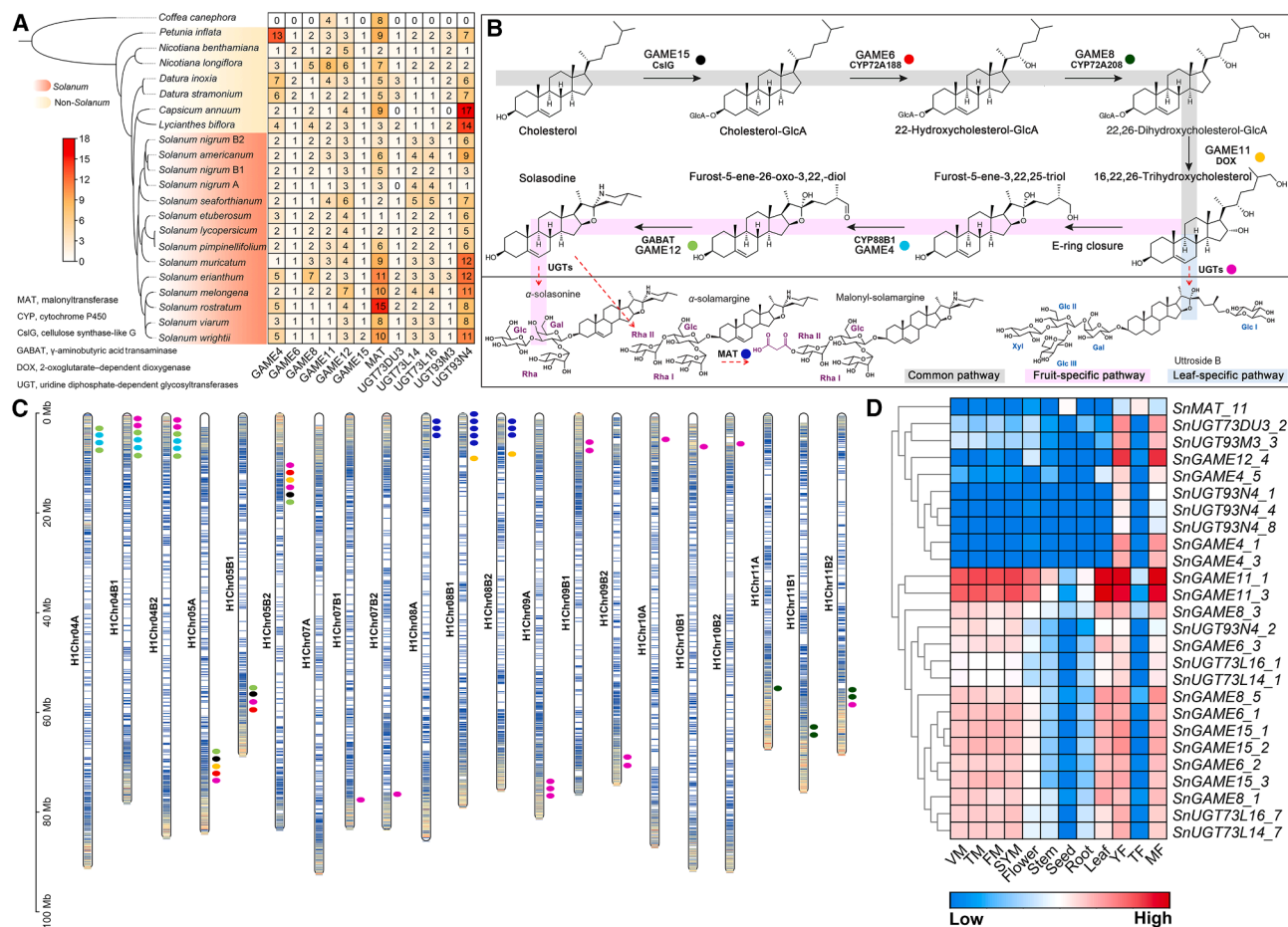


Figure 6. Identify genes associated with SGA biosynthesis in *S. nigrum*

(A) Comparative analysis of SGA biosynthesis-related family members in the Solanaceae genome.

(B) Proposed SGA biosynthesis in *S. nigrum* adapted from the framework established by Lucier et al., Akiyama et al., and Boccia et al. ^{13,34,35}

(C) The distribution of SGA biosynthesis genes in the chromosomes of *S. nigrum*.

(D) Heatmaps showing the transcript accumulation patterns of putative 24 key genes linked to SGA production in *S. nigrum* as determined by RNA sequencing. FM, floral meristem; L, leaf; MF, mature fruit; SYM, sympodial shoot meristem; TF, transition fruit; TM, transition meristem; VM, vegetative meristem; and YF, young fruit.

(Figures 6C, S21–S32). We also further performed comparative synteny analysis of these genes between *S. nigrum* and tomato and between *S. nigrum* and potato, which suggested that the expansion of the *GAME* gene family in *S. nigrum* is primarily attributed to its hexaploid genome (Figure S33). This characteristic highlights the central role of WGT events in the expansion of the *S. nigrum* SGA-related gene family, while also implying subsequent gene loss or neo-functionalization (Figures S34–S45), thereby shaping the current genomic landscape.

Analysis of these gene-expression patterns indicates that multiple coding genes often encode the same enzyme, playing complementary or redundant roles in the evolution of *S. nigrum* and thereby enhancing the robustness and adaptability of its metabolic networks (Figures S34–S45). In addition, unlike the extensively evolved genes of ancient polyploids, the duplicated genes involved in SGA biosynthesis in *S. nigrum* may still be undergoing sub-functionalization or neo-functionalization. To investigate their

evolutionary fate, we calculated the ratio of K_a to K_s substitution rates for homologous gene pairs. The resulting K_a/K_s ratios spanned a wide range from 0.031 to 1.127, indicating that these genes have experienced diverse selective pressures. Our analysis revealed clear examples of the three classic models of gene duplication. The *SnMAT_2/SnMAT_9* pair ($K_a/K_s \approx 1.127$, >1) shows evidence of positive selection, suggesting neofunctionalization (Data S8). The *SnGAME12_1/SnGAME12_9* pair ($K_a/K_s \approx 0.723$) exemplifies subfunctionalization; its relaxed purifying selection is supported by neatly partitioned expression, with *SnGAME12_1* expressed mainly in fruits and *SnGAME12_9* in leaves. Finally, the *SnGAME6_1/SnGAME6_3* pair is under strong purifying selection ($K_a/K_s \approx 0.031$). This high sequence conservation, combined with broad, overlapping expression in most tissues, suggests functional redundancy or a gene dosage effect.

Research indicates that SGAs in *S. nigrum* are primarily synthesized in its fruits and leaves.⁴⁰ The study identified 26 key genes

exhibiting strong or specific expression in these two tissue types, including *SnGAME4_1/3/5*, *SnGAME6_1/2/3*, *SnGAME8_1/3/5*, *SnGAME11_1/3*, *SnGAME12_4*, *SnGAME15_1/2/3*, *SnMAT_11*, *SnUGT73DU3_2*, *SnUGT73L16_7/1*, *SnUGT93M3_3*, *SnUGT73L14_1/7*, and *SnUGT93N4_1/2/4/8* (Figure 6D). To further validate the functions of these genes, we correlated our findings with a comprehensive metabolite analysis of *S. nigrum* fruits conducted by Zhao et al. (2023).⁴¹ Their study revealed that fruit maturation involves a significant chemical transformation of SGAs, where immature fruits contain simpler forms and mature fruits accumulate highly derivatized molecules. This metabolic shift shows a remarkable correlation with our gene expression data (Figure 6D). For instance, the accumulation of malonylated and poly-glycosylated SGAs in ripe fruit, as reported by Zhao et al. (2023),⁴¹ aligns perfectly with the specific, high-level expression of *SnMAT_11* and several *UGTs* (e.g., *SnUGT93M3_3*) in mature fruit samples. Furthermore, the complex series of aglycone oxidations (hydroxylation and carboxylation) described by Zhao et al. (2023)⁴¹ can be mapped to the differential expression of our candidate *GAME* genes. For example, genes such as *SnGAME11_1* and *SnGAME8_3*, which are highly expressed in young fruit and leaves, are likely candidates for catalyzing foundational oxidative steps. Overall, integrated transcriptomic and metabolomic analyses identified a core 26-gene module governing the biosynthesis of SGAs in *S. nigrum*, providing a definitive genetic basis for the pathway's reconstruction in heterologous hosts like tobacco.

DISCUSSION

Repetitive sequences, nearly identical subgenomes, large genomes, and high heterozygosity in polyploids, especially autopolyploids, impede haplotype-resolved, chromosome-level assemblies, limiting detailed studies of allele-specific expression, genomic imprinting, and heterosis.^{42,43} Chromosome-level assemblies of diploid genomes are now common, but haplotype-resolved assemblies in autopolyploid plants are still uncommon,^{44,45} reporting only for a few species such as Jerusalem artichoke, potato, oil-Camellia, and sugarcane.^{46–50} These efforts usually rely on high-quality genetic maps from selfed populations to assign contigs to haplotypes or on closely related diploid reference genomes with conserved chromosomal structures to sort contigs into homologous groups. Uniquely mapped Hi-C reads are then processed with ALLHiC, through trimming, segmentation, and optimization, to anchor each group to pseudochromosomes, producing accurate chromosome-level haplotype assemblies.⁵¹ Although a draft genome of allo-autopolyploid hexaploid *S. nigrum* exists,²⁴ its complexity has impeded chromosome-level assembly. Using PacBio circular consensus sequencing and iterative optimization, we phased haplotypes and achieved a chromosome-scale assembly, providing a robust foundation for functional genomics and resource utilization.

Allo-autopolyploid plants are excellent models for investigating hybrid speciation and reticulate evolution.^{46,52} Evidence from phylogenetic relationships, karyotype analysis, Smudgeplot results, and K-based clustering analysis suggests that *S. americanum* is the ancestor of the existing hexaploid *S. nigrum*:

this species first underwent a homeotic doubling, followed by hybridization with another diploid species about 8.8 MYA, leading to the formation of an allo-autopolyploid hexaploid genome. While early cytological studies suggested that hexaploid *S. nigrum* is an allo-autopolyploid species,¹⁰ this study provides more compelling evidence based on genome-wide analyses. Through multiple lines of evidence, we not only confirmed *S. americanum* as one of its progenitors but also reliably estimated the time window of its hybridization event. In addition, this study further reveals that hexaploid *S. nigrum* exhibits large-scale chromosomal rearrangements compared to its progenitor *S. americanum*, indicating significant genomic structural remodeling. This phenomenon suggests that chromosomal rearrangement may play a crucial role in the formation of *S. nigrum* polyploids and subsequent adaptive evolution, providing a new perspective for elucidating the genomic evolution mechanisms of *Solanum* plants.

SGAs are nitrogen-containing metabolites produced by many *Solanum* species, including *S. lycopersicum*, *S. tuberosum*, *S. melongena*, and *S. nigrum*.^{12–16} SGAs defend plants against pathogens, pests, and herbivores, yet some, such as α -solanine and α -chaconine, are toxic and bitter, making them anti-nutritional for humans.^{12,16,36,53} In contrast, several SGAs, including α -tomatine in *S. lycopersicum* and α -solanine in *S. tuberosum* and *S. melongena*, exhibit notable pharmacological effects, including anticancer, antibacterial, anti-inflammatory, antiviral, and antipyretic activities.^{12,37,47} In recent years, *S. nigrum* has garnered attention for its potent anti-tumor and antiproliferative effects on the liver, cervical, lung, breast, and colorectal cancer cell lines, effects largely attributed to the high concentrations of the SGAs' α -solanine and α -chaconine in its green fruits.^{47,54,55} Despite some *GAME* enzymes being characterized, attempts to reconstitute SGA biosynthesis in heterologous model plants have failed, suggesting the presence of undiscovered genes. To bridge this gap, we generated the first high-quality, haplotype-resolved genome of *S. nigrum*, highlighting the evolutionary histories of SGA pathway-related genes and key members likely involved in SGA biosynthesis. The previously cloned number of genes encoding enzymes related to *S. nigrum* was significantly lower than expected for this hexaploid species, likely due to limitations in PCR cloning techniques.^{13–15} This study analyzed the hexaploid genome and identified all gene copies potentially involved in the SGA biosynthesis in *S. nigrum*, while also elucidating their evolutionary trajectories through multiple genome polyploidization events. Additionally, it identified actively expressed gene copies associated with SGA metabolism, specifically core genes crucial for SGA synthesis. These findings enhance our understanding of SGA biosynthesis in *S. nigrum* and provide valuable genetic resources for future research and industrial production of SGAs. Furthermore, the high-quality reference genome developed in this study will facilitate gene mining, breeding improvements, and industrial applications of this multi-functional crop.

In summary, we have, for the first time, completed a high-quality, haplotype-resolved genome assembly of the hexaploid *S. nigrum*, providing a crucial foundation for understanding Solanaceae genome evolution and the impact of chromosomal structural variations on rapid speciation. Through systematic comparative genomic analysis, we confirmed that the diploid

S. americanum is one of the parental species of the *S. nigrum* genome, further elucidating its polyploid origin. Integrating evolutionary information with transcriptome data from different tissues, we identified a set of core genes potentially involved in SGA biosynthesis, offering a valuable reference for reconstructing the SGA pathway in model plants such as tobacco. The establishment of this reference genome will not only facilitate the application of molecular marker-assisted breeding, genome editing, and other technologies in *S. nigrum* but also provide methodological insights for assembling other large polyploid genomes with complex origins and high heterozygosity. Overall, the findings of this study pave new avenues for genetic improvement and functional gene analysis of *S. nigrum* and other Solanaceae plants.

Limitations of the study

This study presents a high-quality, haplotype-resolved, chromosome-level genome assembly for *S. nigrum*. However, the assembly contains five gaps due to highly repetitive sequences and homologous regions, necessitating future telomere-to-telomere assembly efforts to fully elucidate these complex genomic regions. Furthermore, while analysis suggests *S. americanum* as a likely progenitor of *S. nigrum*, the origin of the second parental species remains undetermined. Although we identified core SGA biosynthetic genes, we lack functional validation; future work should reconstruct the pathway in model plants using these genes to validate it and enable heterologous SGA production.

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Yunpeng Cao (xfcypeng@126.com).

Materials availability

All unique reagents generated in this study are available from the [lead contact](#) with a completed material transfer agreement.

Data and code availability

- The raw sequence data reported in this paper have been deposited at the Genome Sequence Archive (GSA) database in the National Genomics Data Center of China National Center for Bioinformation/Beijing Institute of Genomics under the project number PRJCA046083 (<https://ngdc.cncb.ac.cn/bioproject/browse/PRJCA046083>). All accession numbers are listed in the [key resources table](#).
- This paper does not report original code.
- Additional information is available and will be shared by the [lead contact](#) upon request.

ACKNOWLEDGMENTS

This research was funded by the Guangxi "Bagui Young Talents" Special Fund, the Guangxi "Qingmiao Talent" Program, the National Natural Science Foundation of China (grant no. 32101486), the CNTC Technology Project (grant no. 110202201012 [JY-12], 110202401009 [JY-09], & 110202404004), and the Beijing Life Science Academy Project (grant no. 2023200CC0270).

AUTHOR CONTRIBUTIONS

Y.C. drafted the initial manuscript based on the planned content. Y.C., B.K., Z.Z., and W.L. contributed information and suggestions. Y.C. and X.L. reviewed and edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

The author affirms that this work was prepared independently, without the utilization of artificial intelligence tools. Consequently, the author assumes full and unequivocal responsibility for the integrity, accuracy, and overall content presented within this publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS](#)
 - Plant materials and growth conditions
- [METHOD DETAILS](#)
 - Sample preparation and sequencing
 - Genome size estimation
 - Genome assembly
 - Genome annotation
 - Quality assessment of the assembly and annotation
 - Phylogenetic analysis
 - Analysis of WGD events and karyotype
 - Sub-/neo-functionalization analyses
 - Identification of core steroidal glycoalkaloids (SGAs) biosynthesis genes
 - RNA extracted and RNA-seq analysis
- [QUANTIFICATION AND STATISTICAL ANALYSIS](#)

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2026.117037>.

Received: September 29, 2025

Revised: December 16, 2025

Accepted: February 3, 2026

Published: February 25, 2026

REFERENCES

1. Zhang, F., Qiu, F., Zeng, J., Xu, Z., Tang, Y., Zhao, T., Gou, Y., Su, F., Wang, S., Sun, X., et al. (2023). Revealing evolution of tropane alkaloid biosynthesis by analyzing two genomes in the Solanaceae family. *Nat. Commun.* **14**, 1446.
2. Wang, X., Gao, L., Jiao, C., Stravrovadis, S., Hosmani, P.S., Saha, S., Zhang, J., Mainiero, S., Strickler, S.R., Catala, C., et al. (2020). Genome of *Solanum pimpinellifolium* provides insights into structural variants during tomato breeding. *Nat. Commun.* **11**, 5817. <https://doi.org/10.1038/s41467-020-19682-0>.
3. Jabamalai, A., Priatama, R.A., Heo, J., and Park, S.J. (2019). Medicinal metabolites with common biosynthetic pathways in *Solanum nigrum*. *Plant Biotechnol. Rep.* **13**, 315–327. <https://doi.org/10.1007/s11816-019-00549-w>.
4. Saleem, T.M., Chetty, C., Ramkanth, S., Alagusundaram, M., Gnanaprakash, K., Rajan, V.T., and Angalaparameswari, S. (2009). *Solanum nigrum* Linn.-A review. *Pharmacogn. Rev.* **3**, 342.
5. Sawai, S., Ohyama, K., Yasumoto, S., Seki, H., Sakuma, T., Yamamoto, T., Takebayashi, Y., Kojima, M., Sakakibara, H., Aoki, T., et al. (2014). Sterol Side Chain Reductase 2 Is a Key Enzyme in the Biosynthesis of Cholesterol, the Common Precursor of Toxic Steroidal Glycoalkaloids

- in Potato. *Plant Cell* 26, 3763–3774. <https://doi.org/10.1105/tpc.114.130096>.
6. Jansing, J., Sack, M., Augustine, S.M., Fischer, R., and Bortesi, L. (2019). CRISPR/Cas9-mediated knockout of six glycosyltransferase genes in *Nicotiana benthamiana* for the production of recombinant proteins lacking β -1,2-xylose and core α -1,3-fucose. *Plant Biotechnol. J.* 17, 350–361. <https://doi.org/10.1111/pbi.12981>.
7. Li, J., Stoddard, T.J., Demorest, Z.L., Lavoie, P.-O., Luo, S., Clasen, B.M., Cedrone, F., Ray, E.E., Coffman, A.P., Daulhac, A., et al. (2016). Multiplexed, targeted gene editing in *Nicotiana benthamiana* for glyco-engineering and monoclonal antibody production. *Plant Biotechnol. J.* 14, 533–542. <https://doi.org/10.1111/pbi.12403>.
8. Mercx, S., Smargiasso, N., Chaumont, F., De Pauw, E., Boutry, M., and Navarre, C. (2017). Inactivation of the β (1,2)-xylosyltransferase and the α (1,3)-fucosyltransferase genes in *Nicotiana tabacum* BY-2 Cells by a Multiplex CRISPR/Cas9 Strategy Results in Glycoproteins without Plant-Specific Glycans. *Front. Plant Sci.* 8, 403.
9. Gao, Y., Fan, Z.-Q., Zhang, Q., Li, H.-L., Liu, G.-S., Jing, Y., Zhang, Y.-P., Zhu, B.-Z., Zhu, H.-L., Chen, J.-Y., et al. (2021). A tomato NAC transcription factor, SINAM1, positively regulates ethylene biosynthesis and the onset of tomato fruit ripening. *Plant J.* 108, 1317–1331. <https://doi.org/10.1111/tj.15512>.
10. Rao, M.K. (1971). Cytology of a pentaploid hybrid and genome analysis in *Solanum nigrum* L. *Genetica* 42, 157–164.
11. Nguyen, H.T., Park, S.J., Kang, B.Y., and Kim, H.H. (2023). Comparative triple-color FISH mapping and genome size advances understanding of the cytogenetic diversity in wild *Solanum* species. *Hortic. Environ. Biotechnol.* 64, 811–817.
12. Zhao, D.-K., Zhao, Y., Chen, S.-Y., and Kennelly, E.J. (2021). *Solanum* steroidal glycoalkaloids: structural diversity, biological activities, and biosynthesis. *Nat. Prod. Rep.* 38, 1423–1444.
13. Lucier, R., Kamileen, M.O., Nakamura, Y., Serediuk, S., Barbole, R., Wurlitzer, J., Kunert, M., Heinicke, S., O'Connor, S.E., and Sonawane, P.D. (2024). Steroidal scaffold decorations in *Solanum* alkaloid biosynthesis. *Mol. Plant* 17, 1236–1254.
14. Jozwiak, A., Panda, S., Akiyama, R., Yoneda, A., Umemoto, N., Saito, K., Yasumoto, S., Muranaka, T., Gharat, S.A., Kazachkova, Y., et al. (2024). A cellulose synthase-like protein governs the biosynthesis of *Solanum* alkaloids. *Science* 386, eadq5721.
15. Lezin, E., Papon, N., and Courdavault, V. (2025). A GAME changer in steroidal metabolite biosynthesis. *Plant Commun.* 6, 101201.
16. Friedman, M. (2006). Potato glycoalkaloids and metabolites: roles in the plant and in the diet. *J. Agric. Food Chem.* 54, 8655–8681.
17. Sun, H., Jiao, W.-B., Krause, K., Campoy, J.A., Goel, M., Folz-Donahue, K., Kukat, C., Huettel, B., and Schneeberger, K. (2022). Chromosome-scale and haplotype-resolved genome assembly of a tetraploid potato cultivar. *Nat. Genet.* 54, 342–348. <https://doi.org/10.1038/s41588-022-01015-0>.
18. Sonawane, P.D., Jozwiak, A., Barbole, R., Panda, S., Abebie, B., Kazachkova, Y., Gharat, S.A., Ramot, O., Unger, T., Wizler, G., et al. (2022). 2-oxoglutarate-dependent dioxygenases drive expansion of steroidal alkaloid structural diversity in the genus *Solanum*. *New Phytol.* 234, 1394–1410.
19. Lin, Y., Ye, C., Li, X., Chen, Q., Wu, Y., Zhang, F., Pan, R., Zhang, S., Chen, S., Wang, X., et al. (2023). *quarTeT*: a telomere-to-telomere toolkit for gap-free genome assembly and centromeric repeat identification. *Hortic. Res.* 10, uhad127.
20. Yu, S., Wang, Y., Li, T., Shi, H., Kong, D., Pang, J., Wang, Z., Meng, H., Gao, Y., Wang, X., et al. (2024). Chromosome-scale assembly and gene editing of *Solanum americanum* genome reveals the basis for thermotolerance and fruit anthocyanin composition. *Theor. Appl. Genet.* 137, 15.
21. Zhang, Y., Guo, W., Yuan, Z., Song, Z., Wang, Z., Gao, J., Fu, W., and Zhang, G. (2023). Chromosome-level genome assembly and annotation of the prickly nightshade *Solanum rostratum* Dunal. *Sci. Data* 10, 341.
22. Osborn, T.C., Pires, J.C., Birchler, J.A., Auger, D.L., Chen, Z.J., Lee, H.-S., Comai, L., Madlung, A., Doerge, R.W., Colot, V., et al. (2003). Understanding mechanisms of novel gene expression in polyploids. *Trends Genet.* 19, 141–147.
23. Zhang, J., Zhang, X., Tang, H., Zhang, Q., Hua, X., Ma, X., Zhu, F., Jones, T., Zhu, X., Bowers, J., et al. (2018). Allele-defined genome of the autopolyploid sugarcane *Saccharum spontaneum* L. *Nat. Genet.* 50, 1565–1573.
24. Lee, E.S., Heo, J., Bang, W.Y., Chougule, K.M., Waminal, N.E., Hong, N.T., Kim, M.J., Beak, H.K., Kim, Y.J., Priatama, R.A., et al. (2023). Engineering homoeologs provide a fine scale for quantitative traits in polyploid. *Plant Biotechnol. J.* 21, 2458–2472.
25. Lin, X., Jia, Y., Heal, R., Prokhorchik, M., Sindalovskaya, M., Olave-Achury, A., Makechemu, M., Fairhead, S., Noreen, A., Heo, J., et al. (2023). *Solanum americanum* genome-assisted discovery of immune receptors that detect potato late blight pathogen effectors. *Nat. Genet.* 55, 1579–1588.
26. Ren, R., Wang, H., Guo, C., Zhang, N., Zeng, L., Chen, Y., Ma, H., and Qi, J. (2018). Widespread whole genome duplications contribute to genome complexity and species diversity in angiosperms. *Mol. Plant* 11, 414–428.
27. Liu, Z., Shen, S., Wang, Y., Sun, S., Yu, T., Fu, Y., Zhou, R., Li, C., Cao, R., Zhang, Y., et al. (2024). The genome of *Stephania japonica* provides insights into the biosynthesis of cepharanthine. *Cell Rep.* 43, 113832.
28. Feng, C., Yang, L., Wang, J., Shi, X., and Kang, M. (2025). *Primulina* pan-genome reveals differential gene retention following whole-genome duplications and provides insights into edaphic specialization. *Cell Rep.* 44, 116763.
29. French (2007). The grapevine genome sequence suggests ancestral hexaploidization in major angiosperm phyla. *nature* 449, 463–467.
30. Murat, F., Armero, A., Pont, C., Klopp, C., and Salse, J. (2017). Reconstructing the genome of the most recent common ancestor of flowering plants. *Nat. Genet.* 49, 490–496.
31. Sun, P., Jiao, B., Yang, Y., Shan, L., Li, T., Li, X., Xi, Z., Wang, X., and Liu, J. (2022). WGDl: a user-friendly toolkit for evolutionary analyses of whole-genome duplications and ancestral karyotypes. *Mol. Plant* 15, 1841–1851.
32. Jia, K.H., Wang, Z.X., Wang, L., Li, G.Y., Zhang, W., Wang, X.L., Xu, F.J., Jiao, S.Q., Zhou, S.S., Liu, H., et al. (2022). SubPhaser: a robust allopolyploid subgenome phasing method based on subgenome-specific k-mers. *New Phytol.* 235, 801–809.
33. Kawai, Y., Ono, E., and Mizutani, M. (2014). Evolution and diversity of the 2-oxoglutarate-dependent dioxygenase superfamily in plants. *Plant J.* 78, 328–343.
34. Akiyama, R., Terami, D., Noda, A., Watanabe, B., Umemoto, N., Muranaka, T., Saito, K., Sugimoto, Y., and Mizutani, M. (2025). Two reductases complete steroidal glycoalkaloids biosynthesis in potato. *New Phytol.* 245, 2632–2644.
35. Boccia, M., Kessler, D., Seibt, W., Grabe, V., Rodríguez López, C.E., Grzech, D., Heinicke, S., O'Connor, S.E., and Sonawane, P.D. (2024). A scaffold protein manages the biosynthesis of steroidal defense metabolites in plants. *Science* 386, 1366–1372.
36. Friedman, M. (2015). Chemistry and anticarcinogenic mechanisms of glycoalkaloids produced by eggplants, potatoes, and tomatoes. *J. Agric. Food Chem.* 63, 3323–3337.
37. Al Sinani, S.S.S., and Eltayeb, E.A. (2017). The steroidal glycoalkaloids solamargine and solasonine in *Solanum* plants. *South Afr. J. Bot.* 112, 253–269.

38. Li, Y., and Luo, J. (2025). From steroidal glycoalkaloids to steroidal saponins: Biosynthesis and ecological role in the *Solanum* genus. *Mol. Plant* 18, 22–24.
39. Bai, F., Wu, M., Huang, W., Xu, W., Wang, Y., Zhang, Y., Zhong, Z., Hong, Y., Pirrello, J., Bouzayen, M., and Liu, M. (2025). Removal of toxic steroidal glycoalkaloids and bitterness in tomato is controlled by a complex epigenetic and genetic network. *Sci. Adv.* 11, eads9601.
40. Haas, B.J., Papanicolaou, A., Yassour, M., Grabherr, M., Blood, P.D., Bowden, J., Couger, M.B., Eccles, D., Li, B., Lieber, M., et al. (2013). De novo transcript sequence reconstruction from RNA-seq using the Trinity platform for reference generation and analysis. *Nat. Protoc.* 8, 1494–1512. <https://doi.org/10.1038/nprot.2013.084>.
41. Zhao, W., Yan, T., Huang, X., and Zhang, Y. (2023). Analysis of steroidal glycoalkaloids and their metabolites in *Solanum nigrum* fruits based on liquid chromatography-tandem mass spectrometry and molecular networking. *J. Sep. Sci.* 46, e2200804. <https://doi.org/10.1002/jssc.202200804>.
42. Van de Peer, Y., Ashman, T.-L., Soltis, P.S., and Soltis, D.E. (2021). Polyploidy: an evolutionary and ecological force in stressful times. *Plant Cell* 33, 11–26.
43. Kong, W., Wang, Y., Zhang, S., Yu, J., and Zhang, X. (2023). Recent advances in assembly of complex plant genomes. *Genom. Proteom. Bioinform.* 21, 427–439.
44. Jiang, L., Lin, M., Wang, H., Song, H., Zhang, L., Huang, Q., Chen, R., Song, C., Li, G., and Cao, Y. (2022). Haplotype-resolved genome assembly of *Bletilla striata* (Thunb.) Reichb. f. to elucidate medicinal value. *Plant J.* 111, 1340–1353.
45. Jiang, L., Li, X., Lyu, K., Wang, H., Li, Z., Qi, W., Zhang, L., and Cao, Y. (2025). Rosaceae phylogenomic studies provide insights into the evolution of new genes. *Horticultural Plant Journal* 11, 389–405.
46. Wang, S., Wang, A., Chen, R., Xu, D., Wang, H., Jiang, F., Liu, H., Qian, W., and Fan, W. (2024). Haplotype-resolved chromosome-level genome of hexaploid Jerusalem artichoke provides insights into its origin, evolution, and inulin metabolism. *Plant Commun.* 5, 100767.
47. Bao, Z., Li, C., Li, G., Wang, P., Peng, Z., Cheng, L., Li, H., Zhang, Z., Li, Y., Huang, W., et al. (2022). Genome architecture and tetrasomic inheritance of autotetraploid potato. *Mol. Plant* 15, 1211–1226.
48. Zhang, J., Qi, Y., Hua, X., Wang, Y., Wang, B., Qi, Y., Huang, Y., Yu, Z., Gao, R., Zhang, Y., et al. (2025). The highly allo-autopolyploid modern sugarcane genome and very recent allopolyploidization in *Saccharum*. *Nat. Genet.* 57, 242–253.
49. Zhu, H., Wang, F., Xu, Z., Wang, G., Hu, L., Cheng, J., Ge, X., Liu, J., Chen, W., Li, Q., et al. (2024). The complex hexaploid oil-Camellia genome traces back its phylogenomic history and multi-omics analysis of Camellia oil biosynthesis. *Plant Biotechnol. J.* 22, 2890–2906.
50. Zhang, L., Shi, Y., Gong, W., Zhao, G., Xiao, S., Lin, H., Li, Y., Liao, Z., Zhang, S., Hu, G., et al. (2024). The tetraploid *Camellia oleifera* genome provides insights into evolution, agronomic traits, and genetic architecture of oil *Camellia* plants. *Cell Rep.* 43, 115032.
51. Zhang, X., Zhang, S., Zhao, Q., Ming, R., and Tang, H. (2019). Assembly of allele-aware, chromosomal-scale autopolyploid genomes based on Hi-C data. *Nat. Plants* 5, 833–845.
52. Owens, G.L., Huang, K., Todesco, M., and Rieseberg, L.H. (2023). Re-evaluating homoploid reticulate evolution in *Helianthus* sunflowers. *Mol. Biol. Evol.* 40, msad013.
53. Sonawane, P.D., Heinig, U., Panda, S., Gilboa, N.S., Yona, M., Kumar, S.P., Alkan, N., Unger, T., Bocobza, S., Pliner, M., et al. (2018). Short-chain dehydrogenase/reductase governs steroidal specialized metabolites structural diversity and toxicity in the genus *Solanum*. *Proc. Natl. Acad. Sci. USA* 115, E5419–E5428.
54. Perteau, M., Kim, D., Perteau, G.M., Leek, J.T., and Salzberg, S.L. (2016). Transcript-level expression analysis of RNA-seq experiments with HISAT, StringTie and Ballgown. *Nat. Protoc.* 11, 1650–1667.
55. Gu, X.-Y., Shen, X.-F., Wang, L., Wu, Z.-W., Li, F., Chen, B., Zhang, G.-L., and Wang, M.-K. (2018). Bioactive steroidal alkaloids from the fruits of *Solanum nigrum*. *Phytochemistry* 147, 125–131.
56. Simão, F.A., Waterhouse, R.M., Ioannidis, P., Kriventseva, E.V., and Zdobnov, E.M. (2015). BUSCO: assessing genome assembly and annotation completeness with single-copy orthologs. *Bioinformatics* 31, 3210–3212.
57. Chen, S., Zhou, Y., Chen, Y., and Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics* 34, i884–i890.
58. Marçais, G., and Kingsford, C. (2011). A fast, lock-free approach for efficient parallel counting of occurrences of k-mers. *Bioinformatics* 27, 764–770.
59. Ranallo-Benavidez, T.R., Jaron, K.S., and Schatz, M.C. (2020). GenomeScope 2.0 and Smudgeplot for reference-free profiling of polyploid genomes. *Nat. Commun.* 11, 1432.
60. Zeng, X., Yi, Z., Zhang, X., Du, Y., Li, Y., Zhou, Z., Chen, S., Zhao, H., Yang, S., Wang, Y., and Chen, G. (2024). Chromosome-level scaffolding of haplotype-resolved assemblies using Hi-C data without reference genomes. *Nat. Plants* 10, 1184–1200.
61. Robinson, J.T., Turner, D., Durand, N.C., Thorvaldsdóttir, H., Mesirov, J.P., and Aiden, E.L. (2018). Juicebox.js provides a cloud-based visualization system for Hi-C data. *Cell Syst.* 6, 256–258.e1.
62. Huang, Y., Wang, Z., Schmidt, M.A., Su, H., Xiong, L., and Zhang, J. (2024). DEGAP: Dynamic elongation of a genome assembly path. *Brief. Bioinform.* 25, bbae194.
63. Li, H. (2018). Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* 34, 3094–3100.
64. Flynn, J.M., Hubley, R., Goubert, C., Rosen, J., Clark, A.G., Feschotte, C., and Smit, A.F. (2020). RepeatModeler2 for automated genomic discovery of transposable element families. *Proc. Natl. Acad. Sci. USA* 117, 9451–9457.
65. Han, Y., and Wessler, S.R. (2010). MITE-Hunter: a program for discovering miniature inverted-repeat transposable elements from genomic sequences. *Nucleic Acids Res.* 38, e199.
66. Kapitonov, V.V., and Jurka, J. (2008). A universal classification of eukaryotic transposable elements implemented in Repbase. *Nat. Rev. Genet.* 9, 411–414.
67. Chen, N. (2004). Using Repeat Masker to identify repetitive elements in genomic sequences. *Curr. Protoc. Bioinformatics* 5, 4–10.
68. Gabriel, L., Brúna, T., Hoff, K.J., Ebel, M., Lomsadze, A., Borodovsky, M., and Stanke, M. (2024). BRAKER3: Fully automated genome annotation using RNA-seq and protein evidence with GeneMark-ETP, AUGUSTUS, and TSEBRA. *Genome Res.* 34, 769–777.
69. Bolger, A.M., Lohse, M., and Usadel, B. (2014). Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114–2120.
70. Stanke, M., Keller, O., Gunduz, I., Hayes, A., Waack, S., and Morgenstern, B. (2006). AUGUSTUS: ab initio prediction of alternative transcripts. *Nucleic Acids Res.* 34, W435–W439.
71. Cantalapiedra, C.P., Hernández-Plaza, A., Letunic, I., Bork, P., and Huerta-Cepas, J. (2021). eggNOG-mapper v2: functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Mol. Biol. Evol.* 38, 5825–5829.
72. Ou, S., and Jiang, N. (2018). LTR_retriever: a highly accurate and sensitive program for identification of long terminal repeat retrotransposons. *Plant Physiol.* 176, 1410–1422.
73. Ou, S., and Jiang, N. (2019). LTR_FINDER_parallel: parallelization of LTR_FINDER enabling rapid identification of long terminal repeat retrotransposons. *Mob. DNA* 10, 48.
74. Ellinghaus, D., Kurtz, S., and Willhoeft, U. (2008). LTRharvest, an efficient and flexible software for de novo detection of LTR retrotransposons. *BMC Bioinform.* 9, 18.

75. Rhie, A., Walenz, B.P., Koren, S., and Phillippy, A.M. (2020). Merqury: reference-free quality, completeness, and phasing assessment for genome assemblies. *Genome Biol.* **21**, 245.
76. Emms, D.M., and Kelly, S. (2019). OrthoFinder: phylogenetic orthology inference for comparative genomics. *Genome Biol.* **20**, 238.
77. Suyama, M., Torrents, D., and Bork, P. (2006). PAL2NAL: robust conversion of protein sequence alignments into the corresponding codon alignments. *Nucleic Acids Res.* **34**, W609–W612.
78. Katoh, K., and Standley, D.M. (2013). MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Mol. Biol. Evol.* **30**, 772–780.
79. Minh, B.Q., Schmidt, H.A., Chernomor, O., Schrempf, D., Woodhams, M.D., Von Haeseler, A., and Lanfear, R. (2020). IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Mol. Biol. Evol.* **37**, 1530–1534.
80. Yang, Z. (2007). PAML 4: phylogenetic analysis by maximum likelihood. *Mol. Biol. Evol.* **24**, 1586–1591.
81. Luo, R., Liu, B., Xie, Y., Li, Z., Huang, W., Yuan, J., He, G., Chen, Y., Pan, Q., Liu, Y., et al. (2012). SOAPdenovo2: an empirically improved memory-efficient short-read de novo assembler. *GigaScience* **1**, 18–2217X.
82. Fu, L., Niu, B., Zhu, Z., Wu, S., and Li, W. (2012). CD-HIT: accelerated for clustering the next-generation sequencing data. *Bioinformatics* **28**, 3150–3152.
83. Buchfink, B., Xie, C., and Huson, D.H. (2015). Fast and sensitive protein alignment using DIAMOND. *Nat. Methods* **12**, 59–60.
84. Gabriel, L., Hoff, K.J., Brúna, T., Borodovsky, M., and Stanke, M. (2021). TSEBRA: transcript selector for BRAKER. *BMC Bioinf.* **22**, 1–12.
85. Benson, G. (1999). Tandem repeats finder: a program to analyze DNA sequences. *Nucleic Acids Res.* **27**, 573–580.
86. Qiao, X., Li, Q., Yin, H., Qi, K., Li, L., Wang, R., Zhang, S., and Paterson, A.H. (2019). Gene duplication and evolution in recurring polyploidization-diploidization cycles in plants. *Genome Biol.* **20**, 38.
87. Allaire, J. (2012). RStudio: integrated development environment for R. Boston, MA, <http://www.rstudio.org/>.
88. Mistry, J., Finn, R.D., Eddy, S.R., Bateman, A., and Punta, M. (2013). Challenges in homology search: HMMER3 and convergent evolution of coiled-coil regions. *Nucleic Acids Res.* **41**, e121.
89. Cheng, H., Concepcion, G.T., Feng, X., Zhang, H., and Li, H. (2021). Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm. *Nat. Methods* **18**, 170–175.
90. Chen, C., Wu, Y., Li, J., Wang, X., Zeng, Z., Xu, J., Liu, Y., Feng, J., Chen, H., He, Y., and Xia, R. (2023). TBtools-II: A “one for all, all for one” bioinformatics platform for biological big-data mining. *Mol. Plant* **16**, 1733–1742.
91. Danecek, P., Bonfield, J.K., Liddle, J., Marshall, J., Ohan, V., Pollard, M.O., Whitwham, A., Keane, T., McCarthy, S.A., Davies, R.M., and Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience* **10**, giab008.
92. Zhang, Z. (2022). KaKs_Calculator 3.0: calculating selective pressure on coding and non-coding sequences. *Genom. Proteom. Bioinform.* **20**, 536–540.
93. Cao, Y., Feng, X., Ding, B., Huo, H., Abdullah, M., Hong, J., Jiang, L., Wang, H., Li, R., Cai, Y., et al. (2025). Gap-free genome assemblies of two *Pyrus bretschneideri* cultivars and GWAS analyses identify a CCCH zinc finger protein as a key regulator of stone cell formation in pear fruit. *Plant Commun.* **6**, 101238. <https://doi.org/10.1016/j.xplc.2024.101238>.
94. Kumar, S., Suleski, M., Craig, J.M., Kaspricz, A.E., Sanderford, M., Li, M., Stecher, G., and Hedges, S.B. (2022). TimeTree 5: an expanded resource for species divergence times. *Mol. Biol. Evol.* **39**, msac174.
95. Huang, J., Xu, W., Zhai, J., Hu, Y., Guo, J., Zhang, C., Zhao, Y., Zhang, L., Martine, C., Ma, H., and Huang, C.H. (2023). Nuclear phylogeny and insights into whole-genome duplications and reproductive development of Solanaceae plants. *Plant Commun.* **4**, 100595.
96. Yan, X., Shi, G., Sun, M., Shan, S., Chen, R., Li, R., Wu, S., Zhou, Z., Li, Y., Liu, Z., et al. (2024). Genome evolution of the ancient hexaploid *Platanus* × *acerifolia* (London planetree). *Proc. Natl. Acad. Sci. USA* **121**, e2319679121.
97. Duarte, J.M., Cui, L., Wall, P.K., Zhang, Q., Zhang, X., Leebens-Mack, J., Ma, H., Altman, N., and DePamphilis, C.W. (2006). Expression pattern shifts following duplication indicative of subfunctionalization and neofunctionalization in regulatory genes of Arabidopsis. *Mol. Biol. Evol.* **23**, 469–478.
98. Hunter, S., Apweiler, R., Attwood, T.K., Bairoch, A., Bateman, A., Binns, D., Bork, P., Das, U., Daugherty, L., Duquenne, L., et al. (2009). InterPro: the integrative protein signature database. *Nucleic Acids Res.* **37**, D211–D215.
99. Letunic, I., Khedkar, S., and Bork, P. (2021). SMART: recent updates, new developments and status in 2020. *Nucleic Acids Res.* **49**, D458–D460.
100. Marchler-Bauer, A., Derbyshire, M.K., Gonzales, N.R., Lu, S., Chitsaz, F., Geer, L.Y., Geer, R.C., He, J., Gwadz, M., Hurwitz, D.I., et al. (2015). CDD: NCBI's conserved domain database. *Nucleic Acids Res.* **43**, D222–D226.
101. Price, M.N., Dehal, P.S., and Arkin, A.P. (2009). FastTree: computing large minimum evolution trees with profiles instead of a distance matrix. *Mol. Biol. Evol.* **26**, 1641–1650.
102. Kim, D., Paggi, J.M., Park, C., Bennett, C., and Salzberg, S.L. (2019). Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. *Nat. Biotechnol.* **37**, 907–915.
103. Cao, Y., Fan, T., Wang, L., Zhang, L., and Li, Y. (2023). Large-scale analysis of putative Euphorbiaceae R2R3-MYB transcription factors identifies a MYB involved in seed oil biosynthesis. *BMC Plant Biol.* **23**, 145.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
<i>Solanum nigrum</i> (leaves, seeds, stem, roots, flowers, mature green (young fruit), purple (transition fruit), and black (mature fruit))	This study	N/A
Deposited data		
Genome	This study	https://github.com/xfcpeng/Genome
RNA-seq	National Genomics Data Center	PRJCA046083
Hi-C	National Genomics Data Center	PRJCA046083
HiFi	National Genomics Data Center	PRJCA046083
Whole genome sequencing	National Genomics Data Center	PRJCA046083
RNA-seq of <i>S. physalifolium</i>	European Bioinformatics Institute	ERR634924
RNA-seq of <i>S. ptychanthum</i>	European Bioinformatics Institute	ERR2040626
RNA-seq of <i>S. paucidentis</i>	National Genomics Data Center	SRR28923986
RNA-seq of <i>S. nigrescens</i>	National Genomics Data Center	SRR28389317
RNA-seq of <i>S. enantiophyllum</i>	National Genomics Data Center	SRR28477432
RNA-seq of <i>S. macrotonum</i>	National Genomics Data Center	SRR28891692
RNA-seq of <i>S. longifilamentum</i>	National Genomics Data Center	SRR28908455
RNA-seq of <i>S. douglasii</i>	National Genomics Data Center	SRR18909634
RNA-seq of <i>S. furcatum</i>	National Genomics Data Center	SRR18909627
RNA-seq of <i>S. sinuatirecurvum</i>	National Genomics Data Center	SRR18909653
RNA-seq of <i>S. didymum</i>	National Genomics Data Center	SRR28477057
RNA-seq of <i>S. laciniatum</i>	National Genomics Data Center	SRR18859090
RNA-seq of <i>S. guineense</i>	National Genomics Data Center	SRR18909682
RNA-seq of <i>S. ferox</i>	National Genomics Data Center	SRR18909689
RNA-seq of <i>S. diphyllum</i>	National Genomics Data Center	SRR18909673
RNA-seq for meristem stages (VM, TM, FM, SYM)	National Center for Biotechnology Information	PRJNA946382
Software and algorithms		
BUSCO	Simão et al. ⁵⁶	https://busco.ezlab.org/
fastp	Chen et al. ⁵⁷	https://github.com/OpenGene/fastp
SubPhaser	Jia et al. ³²	https://github.com/zhangrengang/SubPhaser
Jellyfish	Marçais and Kingsford ⁵⁸	https://github.com/jamesturk/jellyfish
GenomeScope	Ranallo-Benavidez et al. ⁵⁹	https://github.com/schatzlab/genomescope
HapHiC	Zeng et al. ⁶⁰	https://github.com/zengxiaofei/HapHiC
Juicebox	Robinson et al. ⁶¹	https://github.com/aidenlab/Juicebox
DEGAP	Huang et al. ⁶²	https://github.com/Jianwei-Zhang/DEGAP
Minimap2	Li ⁶³	https://github.com/lh3/minimap2
RepeatModeler	Flynn et al. ⁶⁴	https://github.com/Dfam-consortium/RepeatModeler
MITE-Hunter	Han and Wessler ⁶⁵	https://github.com/jburnette/MITE-Hunter
Rebase	Kapitonov and Jurka ⁶⁶	https://www.girinst.org/rebase/

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
RepeatMasker	Chen ⁶⁷	https://github.com/Dfam-consortium/RepeatMasker
BRAKER3	Gabriel et al. ⁶⁸	https://github.com/Gaius-Augustus/BRAKER
Trimmomatic	Bolger et al. ⁶⁹	https://github.com/usadellab/Trimmomatic
HISAT2	Pertea et al. ⁵⁴	https://github.com/DaehwanKimLab/hisat2
StringTie	Pertea et al. ⁵⁴	https://github.com/gpertea/stringtie
GeneMark	Stanke et al. ⁷⁰	https://github.com/gatech-genemark/GeneMark-ETP
eggNOG-mapper	Cantalapiedra et al. ⁷¹	http://eggno-mapper.embl.de/
LTR retriever	Ou and Jiang ⁷²	https://github.com/oushujun/LTR_retriever
LTR_FINDER	Ou and Jiang ⁷³	https://github.com/xzhub/LTR_Finder
LTRHarvest	Ellinghaus et al. ⁷⁴	https://www.zbh.uni-hamburg.de/forschung/gi/software/ltrharvest.html
Mercury	Rhie et al. ⁷⁵	https://github.com/marbl/mercury
OrthoFinder	Emms and Kelly ⁷⁶	https://github.com/davidemms/OrthoFinder
PAL2NAL	Suyama et al. ⁷⁷	https://www.bork.embl.de/pal2nal/
MAFFT	Katoh and Standley ⁷⁸	https://mafft.cbrc.jp/alignment/software/
IQ-TREE	Minh et al. ⁷⁹	http://www.iqtree.org/
PAML	Yang ⁸⁰	https://github.com/HLTCHKUST/PAML
SOAPdenovo2	Luo et al. ⁸¹	https://github.com/aquaskyline/SOAPdenovo2
Trinity	Haas et al. ⁴⁰	https://github.com/wlanjie/trinity
TransDecoder	NA	https://github.com/TransDecoder/TransDecoder
CD-HIT	Fu et al. ⁸²	http://weizhongli-lab.org/cd-hit/
Diamond	Buchfink et al. ⁸³	https://github.com/python-diamond/Diamond
TSEBRA	Gabriel et al. ⁸⁴	https://github.com/Gaius-Augustus/TSEBRA
Tandem Repeats Finder	Benson ⁸⁵	https://github.com/Benson-Genomics-Lab/TRF
WGDI	Sun et al. ³¹	https://github.com/SunPengChuan/wgdi
DupGen_finder	Qiao et al. ⁸⁶	https://github.com/qiao-xin/DupGen_finder
RStudio	Allaire ⁸⁷	https://github.com/rstudio/rstudio
HMMER	Mistry et al. ⁸⁸	http://hmmer.org/
Hifiasm	Cheng et al. ⁸⁹	https://github.com/chhyllp123/hifiasm
quarTeT	Lin et al. ¹⁹	https://github.com/aaranyue/quarTeT
TBtools-II	Chen et al. ⁹⁰	https://github.com/CJ-Chen/TBtools-II
SAMtools	Danecek et al. ⁹¹	https://github.com/samtools/samtools
KaKs_Calculator	Zhang et al. ⁹²	https://ngdc.cnbc.ac.cn/biocode/tools/BT000001

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Plant materials and growth conditions

S. nigrum was cultivated at the Research Institute of Technology Center, China Tobacco Hunan Industrial Co., Ltd., in Changsha, Hunan, China. The plants were cultivated in a greenhouse under long-day conditions (40–60% relative humidity, 18°C–20°C

temperature, 8–10 h of darkness, 26°C–28°C temperature, and 14–16 h of light), supplemented by 200-watt halogen lamps. Seeds were sown directly in 50-cell plastic flats and later transferred to pots after four weeks. A standard fertilizer regimen and drip irrigation were employed for all plants.

METHOD DETAILS

Sample preparation and sequencing

The high-quality genomic DNA was extracted by the Qiagen Genomic DNA Kit according to the standard protocol of Qiagen (<https://www.qiagen.com>). The SMRTbell libraries were constructed using 15-kb preparation solutions based on the manufacturer's guidelines of PacBio (<https://www.pacb.com>). The 15-μg genomic DNA sample was sheared using g-TUBEs (Covaris, <https://www.covaris.com>) for HiFi sequencing on a PacBio Sequel II instrument with Sequel II Binding Kit 2.0 and Sequencing Primer V2 in GrandOmics, following the standard protocol. For Illumina sequencing (<https://www.illumina.com>), the Hi-C library underwent 150-bp paired-end sequencing on a DNBSEQ (MGISEQ-2000) sequencing platform (BGI).

Genome size estimation

Sequencing data for the *S. nigrum* genome survey were generated on the BGI DNBSEQ platform. Clean reads were generated by filtering raw data with fastp v2.0,⁵⁷ which eliminates adaptor sequences and reads containing more than 10% bases with $Q \leq 20$ or over 1% N content. A total of 130.3 Gb of clean data were then used for k-mer analysis to estimate genome size utilizing SubPhaser v1.2.5³² and Jellyfish v2.2.6.⁵⁸ GenomeScope 2.0 was used to estimate the repetition rate, heterozygosity rate, and genome size.⁵⁹

Genome assembly

The genome of *S. nigrum* was assembled as follows: a) *De novo* phased assemblies were built with Hifiasm v0.19.8 under default settings, combining Hi-C and HiFi reads⁸⁹; b) Hi-C data then guided HapHiC v1.0.6 to scaffold contigs and eliminate redundant unanchored sequences⁶⁰; c) The resulting scaffolds were reviewed and refined in Juicebox v1.11.08.⁶¹ DEGAP was used to close the assembled genome gaps via a cyclic elongation method, as described by Cao et al. (2025).⁹³ Gaps were closed by iteratively extending one edge with HiFi reads. Reads were first mapped to a 60 kb edge using Minimap2 (-Y -ax asm20).⁶³ Those showing high identity and adequate overhang were assembled with Hifiasm, and the best-aligning contig or read was appended. If the extension reached the opposite boundary, the gap was closed; otherwise, the new edge was used for the next round. This cycle continued until a long, high-identity alignment fully bridged the gap.⁶²

Genome annotation

Repetitive sequences were annotated in two steps. First, a *de novo* repeat library was generated using RepeatModeler v1.0.4⁶⁴ and MITE-Hunter v.11-2011⁶⁵ and merged with Repbase v21.12 (<https://www.girinst.org/repbase/>)⁶⁶ to create the final database. Then, RepeatProteinMask v4.0.7 and RepeatMasker v4.0.7⁶⁷ searched the genome against this library. Tandem repeats were detected using Tandem Repeats Finder (TRF, v4.10.0rc2)⁸⁵ with parameters “2 7 7 80 10 50 500”. All detected repeats were masked before genome annotation, conducted using BRAKER3 on the soft-masked genome.⁶⁸ First, Trimmomatic v 0.39 was used for quality filtering and trimming of RNA-seq reads.⁶⁹ Next, HISAT2 v 2.2.1 was employed to map the RNA-seq libraries.⁵⁴ The resulting mapping data, along with protein sequences from five genomes (*A. thaliana*, *N. attenuata*, *S. lycopersicum*, *S. stenotomum*, and *S. tuberosum*), were input into the BRAKER pipeline.⁶⁸ This pipeline employs StringTie v 2.2.0 for RNA-seq read assembly,⁵⁴ followed by gene prediction and training using AUGUSTUS v 3.8.0 and GeneMark v 1.0.⁷⁰ The resulting gene sets are integrated with TSEBRA v 1.0.3.⁸⁴ Finally, the eggNOG-mapper v 5.0 was used to generate functional annotations,⁷¹ which identify potential functions through homology with sequences in the KEGG, Gene Ontology (GO), Pfam, and Swiss-Prot databases.

Quality assessment of the assembly and annotation

We used the BUSCOs to evaluate the completeness of both genome sequences and annotated proteins with the embryophyte_odb10 dataset.⁵⁶ For genome quality assessment, Python v3.8.5 was used to execute the BUSCOs with the command line parameters being ‘busco3.0.2/bin/run busco -i genome.fa -m geno -l embryophyta odb10 -o out name -c 15’. For annotation quality assessment, Python 3.8.5 was used to execute the BUSCOs with the command line parameters being ‘busco3.0.2/bin/run busco -i pep.fa -m prot -l embryophyta odb10 -o out name -c 15’.

Using the LTR retriever tool, the LTR assembly index (LAI) was calculated to assess the completeness of the repeat space.⁷² In the process, the wrapper LTR_FINDER_parallel was used to run LTR_FINDER v1.1 with the parameters of ‘-threads 20 -size 1,000,000 -time 300 -harvest_out -seq genome.fa’.⁷³ LTRHarvest v1.6.1 was run using the parameters of ‘-seqids yes -seed 20 -vic 10 -similar 85 -motifmis 1 -motif TGCA -maxtsd 6 -mintsd 4 -maxlenltr 7000 -index genome.fa -minlenltr 100’.⁷⁴ Subsequently, LTR_retriever v2.9.0 was run using the default parameters.⁷² The Merquy v1.3 was used to evaluate the base accuracy (QV) of the genome with default settings.⁷⁵

Phylogenetic analysis

The OrthoFinder v2.5.5 was used to obtain OGs within the *S. nigrum* genome, as well as those of 10 other species (*Coffea canephora*, *Petunia inflata*, *N. benthamiana*, *N. longiflora*, *Datura innoxia*, *D. stramonium*, *Capsicum annuum*, *Lycianthes biflora*, *S. americanum*, *S. seaforthianum*, *S. etuberosum*, *S. lycopersicum*, *S. pimpinellifolium*, *S. muricatum*, *S. erianthum*, *S. melongena*, *S. rostratum*, *S. viarum*, and *S. wrightii*).⁷⁶ For each single-copy OG, nucleotide alignments were generated using PAL2NAL v 14.0,⁷⁷ and amino acid alignments were carried out using MAFFT v7.⁷⁸ IQ-TREE v2.4.0 was used to generate an ML phylogeny from the concatenated alignments of all single-copy genes.⁷⁹ Furthermore, MCMCTREE in PAML v4.9 was used to estimate species divergence dates.⁸⁰ The following criteria were used to choose the calibration points: 72.5–97.6 MYA between *C. canephora* and *P. inflata*, 11.2–45.9 MYA between *D. innoxia* and *C. annuum*, and 16.1–22.7 MYA between *L. biflora* and *S. viarum*.⁹⁴

To better identify extant close relatives of *S. nigrum* that most closely resemble its true ancestor, we obtained additional diploid transcriptome or genome datasets from public databases. All RNA-seq data were assembled with SOAPdenovo2 v2.40⁸¹ and Trinity v2.15.2.⁴⁰ TransDecoder v3.0 was used to predict the coding sequences for each gene/transcript with default parameters,⁴⁰ and CD-HIT v4.8.1 was used to remove redundant contigs at a sequence identity threshold of 0.95.⁸² OrthoFinder v2.5.5 was used to infer low-copy OGs in *S. nigrum* and its close relatives: *S. didymium*, *S. aciniatum*, *S. guineense*, *S. seaforthianum*, *S. dulcamara*, *S. physalifolium*, *S. ptychanthum*, *S. americanum*, *S. paucidentis*, *S. enantiophyllum*, *S. macrotonum*, *S. nigrescens*, *S. furcatum*, *S. douglasii*, *S. sinuatirecurum*, and *S. longifilamentum*, as described by Huang et al. (2023).⁹⁵ IQ-TREE v2.4.0 was used to generate an ML phylogeny from the concatenated alignments of low-copy OGs.⁷⁹

Analysis of WGD events and karyotype

Diamond v2.1.12 with a cut-off E-value of 1e-5 was utilized for all-against-all blastp comparisons.⁸³ Syntenic genes in collinear regions were identified using WGDI v0.74, based on the top five non-self hits.³¹ The yn00 tool from PAML v4.9 was employed to calculate the ratio of Ka to Ks for all paralogous genes and their orthologs, as well as the Ks values for syntenic genes, using default parameters.⁸⁰ Genes were identified as under positive selection if a Likelihood Ratio Test (LRT) was significant ($p < 0.05$) with Ka/Ks > 1 using KaKs_Calculator v3.0.⁹² WGDI v0.74 was used to reconstruct the projections of the AEK following the methodology of Sun et al. (2022).³¹ Diamond v2.1.12 was used to compare the protein sequences of the eight Solanaceae species with the AEK using the options “-num_alignments 20 -outfmt 6 -evalue 1e-5”.⁸³ Chromosome information and gene locations for WGDI were obtained using the generate_conf.py script (<https://github.com/xuzhougeng/>). Second, the syntenic blocks were filtered, integrated, and validated with the “-bi -c -bk” options in WGDI, and collinear genes between the current genomes and the two ancestral karyotypes were identified using the “-ici” parameter. The mapping from AEK to the karyotype was achieved using WGDI with the “-km” parameter. Finally, the ancestral karyotype projections were visualized using WGDI to create homologous dot plots between the two ancestral karyotypes and the current genomes. To assess the diversity in karyotype evolution and chromosomal rearrangement, we compared the protein sequences of *S. nigrum* with those from other Solanaceae genomes using Diamond v2.1.12.⁸³

Sub-/neo-functionalization analyses

The gene copies present in triplicate were identified, following the DupGen_finder pipeline, as previously described by.⁸⁶ To compare gene expression across the subgenomes of the haploid genome, we calculated the annual expression profile, which refers to the gene expression of genes in different tissues. We analyzed overall mean expression across tissues (G effect), tissue-specific mean differences linked to gene copy number (R effect), and their interaction (G × R effect), as previously described by Yan et al. (2024) and Duarte et al. (2006).^{96,97} Expression patterns were analyzed by split-plot ANOVA of duplicate pairs in RStudio v2025.05.1–513.⁸⁷ A significant G × R interaction indicates that genes from subgenomes A, B1, and B2 differ in expression across time, producing complementary patterns termed sub-/neo-functionalization.⁹⁷

Identification of core steroidal glycoalkaloids (SGAs) biosynthesis genes

The biosynthesis of SGA in hexaploid *S. nigrum* requires further investigation compared to other Solanaceae species due to the challenges posed by its high ploidy and heterozygosity, which hinder the creation of a high-quality genome. Using a high-quality, haplotype-resolved genome, this study systematically identifies genes associated with the biosynthesis of SGAs in *S. nigrum*. Initially, HMM files related to oxidase-mediated oxidation/hydroxylation, UGTs, and CYPs were sourced from the InterPro website.⁹⁸ HMMER v3.4 was then employed to identify genes within these families based on the HMM files.⁸⁸ Finally, the identified gene members were validated for characteristic conserved domains through InterPro, SMART, and CDD databases.^{98–100} Subsequently, we employ experimentally validated SGA-related gene sequences, supported by forward genetic evidence, and conduct evolutionary analyses using FastTree v1.9 or OrthoFinder v2.5.5 to identify all potential genes involved in SGA biosynthesis in Solanaceae plants, thereby ensuring the accuracy and reliability of our findings.^{76,79,101}

RNA extracted and RNA-seq analysis

RNA-seq samples were collected from leaves, seeds, stems, roots, flowers, mature green (young fruit), purple (transition fruit), and black (mature fruit). Among them, the fourth true leaves (excluding cotyledons), roots, flowers, and stems were collected at 25 days after sowing (DAS), and fruits were sampled at three distinct stages: mature green (young fruit), purple (transition fruit), and black (mature fruit). Each fruit-stage biological replicate comprised more than 15 fruits. All samplings occurred between

9:00 and 11:00 a.m., followed by immediate flash-freezing in liquid nitrogen. All tissue samples were homogenized using a Mixer Mill MM400 (Retsch GmbH, Haan, Germany) with 3-mm tungsten beads. Total RNA was extracted using the RNeasy Pure Plant Plus Kit (Qiagen, Beijing, China), and mRNA was isolated, fragmented, and purified with the TruSeq RNA Library Prep Kit v.2 (Illumina, San Diego, CA, USA). The libraries were sequenced on the BGISEQ platform. Furthermore, publicly available transcriptome data for meristem stages (VM, TM, FM, SYM), staged by plastochron index and days after germination (DAG), were incorporated from the Sequence Read Archive under accession PRJNA946382. All raw RNA-seq data were trimmed using Trimmomatic v0.39.³⁶ Clean reads were mapped against the genome assembly using HISAT2 v2.0.5 with default parameters, guided by the GFF3 file.¹⁰² Then, SAMtools v1.3 was used to filter the mapped BAM files with parameters “-F 256 -q 30 -f 2” to retain only high-quality, correctly mapped read pairs,⁹¹ thereby minimizing the impact of repeated mappings on the expression profiling of highly similar homologous genes, as described by Wang et al. (2024).⁴⁶ Finally, read counts and TPM values were computed using StringTie,¹⁰³ and visual analysis was conducted using TBtools-II v2.3.⁹⁰

QUANTIFICATION AND STATISTICAL ANALYSIS

All details of the statistics applied in this study are provided alongside the respective analysis in the method details section.